

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - cutoff = 0.75	5.9	3.5	0.6	0.6	0.2	89.2
U2 - cutoff = 0.85	15.9	8	1.6	0.5	0.2	73.8
U2 - cutoff = 0.95	30.4	11	1	0.4	0.2	57
U3 - cutoff = 0.75	6.1	3.7	0.7	0.2	0.2	89.1
U3 - cutoff = 0.85	17.3	7.7	1.3	0.3	0.2	73.2
U3 - cutoff = 0.95	33.3	8.9	0.3	0.2	0.2	57.1
Mean administrations						
U2 - cutoff = 0.75	7.8	3	1.9	0.9	0.4	-
U2 - cutoff = 0.85	11.1	3.6	2.1	0.9	0.3	-
U2 - cutoff = 0.95	15.4	3.8	2.1	0.9	0.3	-
U3 - cutoff = 0.75	7.8	3	2	0.9	0.4	-
U3 - cutoff = 0.85	11.1	3.6	2.2	0.9	0.3	-
U3 - cutoff = 0.95	15.4	3.8	2.1	0.9	0.3	-
Mean IPDE administrations						
U2 - cutoff = 0.75	2	1.6	0.7	0.4	0.1	-
U2 - cutoff = 0.85	3.2	1.8	0.7	0.4	0.1	-
U2 - cutoff = 0.95	0.6	2	0.6	0.4	0.1	-
U3 - cutoff = 0.75	2	1.5	0.7	0.4	0.1	-
U3 - cutoff = 0.85	3.2	1.8	0.7	0.4	0.1	-
U3 - cutoff = 0.95	0.6	2	0.6	0.4	0.1	-
Mean trial duration (days)						
U2 - cutoff = 0.75						Days
U2 - cutoff = 0.85						501.6
U2 - cutoff = 0.95						642.5
U3 - cutoff = 0.75						1049.1
U3 - cutoff = 0.85						501.4
U3 - cutoff = 0.95						645
						1050.1
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - cutoff = 0.75	25	14.4	32.6	11.2	3.7	13.1
U2 - cutoff = 0.85	31.7	15	33.2	11.2	3.7	5.2
U2 - cutoff = 0.95	34.8	16.2	32	11.9	3.2	1.9
U3 - cutoff = 0.75	28	15.4	31	10	2.6	13
U3 - cutoff = 0.85	34.3	16.3	31.2	10.2	3	5
U3 - cutoff = 0.95	37.8	17	30.7	10.2	2.5	1.8
Mean administrations						
U2 - cutoff = 0.75	9.5	5	7.2	3.9	2.1	-
U2 - cutoff = 0.85	10.8	5.1	7.2	3.8	2.2	-
U2 - cutoff = 0.95	11.4	5.2	7.2	3.9	2	-
U3 - cutoff = 0.75	9.6	4.9	7.3	3.8	2.2	-
U3 - cutoff = 0.85	10.8	5.1	7.1	3.9	2.1	-
U3 - cutoff = 0.95	11.5	5.1	7.1	3.9	2.1	-
Mean IPDE administrations						
U2 - cutoff = 0.75	2.6	2.4	1.9	1	0.5	-
U2 - cutoff = 0.85	3.1	2.4	1.9	1	0.5	-
U2 - cutoff = 0.95	0.4	2.4	1.4	0.9	0.3	-
U3 - cutoff = 0.75	2.6	2.4	1.9	1	0.6	-
U3 - cutoff = 0.85	3.1	2.4	1.9	1.1	0.5	-
U3 - cutoff = 0.95	0.4	2.4	1.3	0.9	0.3	-
Mean trial duration (days)						
U2 - cutoff = 0.75						Days
U2 - cutoff = 0.85						1061.6
U2 - cutoff = 0.95						1107.9
U3 - cutoff = 0.75						1347
U3 - cutoff = 0.85						1062.4
U3 - cutoff = 0.95						1107.9
						1346.8

Phase I/II Operating Characteristics

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - cutoff = 0.75	7.1	36.4	21.2	8.2	2.6	24.5
U2 - cutoff = 0.85	10.5	39.5	21	8.2	2.7	18.1
U2 - cutoff = 0.95	18.4	40.3	21.4	7.3	2.6	10
U3 - cutoff = 0.75	8.7	40.3	18.5	6.5	1.6	24.4
U3 - cutoff = 0.85	12.6	42.7	18.9	5.7	1.9	18.2
U3 - cutoff = 0.95	20.1	42.6	18.8	6.3	1.9	10.1
Mean administrations						
U2 - cutoff = 0.75	6.3	7.4	6.1	3.6	2	-
U2 - cutoff = 0.85	7.8	7.5	6.1	3.6	2	-
U2 - cutoff = 0.95	9.7	7.5	6.2	3.7	1.9	-
U3 - cutoff = 0.75	6.3	7.3	6.2	3.5	2	-
U3 - cutoff = 0.85	7.8	7.6	6	3.6	2	-
U3 - cutoff = 0.95	9.7	7.6	6.1	3.6	1.9	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.4	3.2	1.7	1.2	0.6	-
U2 - cutoff = 0.85	2	3.3	1.7	1.1	0.6	-
U2 - cutoff = 0.95	0.2	2.9	1.2	1	0.4	-
U3 - cutoff = 0.75	1.4	3.2	1.7	1.1	0.6	-
U3 - cutoff = 0.85	2	3.3	1.7	1.1	0.6	-
U3 - cutoff = 0.95	0.2	2.9	1.2	1	0.4	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						960.8
U2 - cutoff = 0.85						1011.7
U2 - cutoff = 0.95						1279.4
U3 - cutoff = 0.75						960.8
U3 - cutoff = 0.85						1011.7
U3 - cutoff = 0.95						1281
Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - cutoff = 0.75	4.5	5.9	18.3	43.4	20.4	7.5
U2 - cutoff = 0.85	6.4	7.2	21.6	42.5	19.4	2.9
U2 - cutoff = 0.95	7.3	8.3	21.3	42.1	19.1	1.9
U3 - cutoff = 0.75	7.9	7.4	18	42.4	16.8	7.5
U3 - cutoff = 0.85	10.6	8.8	18.7	41.8	17.2	2.9
U3 - cutoff = 0.95	11.3	8.8	20.2	41.5	16.2	1.9
Mean administrations						
U2 - cutoff = 0.75	4	4	6.3	8	6.5	-
U2 - cutoff = 0.85	4.3	4.2	6.5	8	6.5	-
U2 - cutoff = 0.95	4.5	4.3	6.6	7.9	6.4	-
U3 - cutoff = 0.75	4	4	6.3	8.1	6.5	-
U3 - cutoff = 0.85	4.3	4.1	6.4	8	6.5	-
U3 - cutoff = 0.95	4.6	4.2	6.6	7.9	6.3	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.5	2.7	2	2.6	1.5	-
U2 - cutoff = 0.85	0.6	2.7	2	2.6	1.5	-
U2 - cutoff = 0.95	0.3	2.8	1.4	2.2	0.7	-
U3 - cutoff = 0.75	0.5	2.7	2	2.6	1.5	-
U3 - cutoff = 0.85	0.6	2.7	2	2.6	1.5	-
U3 - cutoff = 0.95	0.3	2.8	1.5	2.2	0.7	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1079.6
U2 - cutoff = 0.85						1097.8
U2 - cutoff = 0.95						1224.1
U3 - cutoff = 0.75						1080.8
U3 - cutoff = 0.85						1097.3
U3 - cutoff = 0.95						1225.8

Phase I/II Operating Characteristics

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - cutoff = 0.75	4.4	35.3	24.7	18.1	10.4	7.1
U2 - cutoff = 0.85	3.7	36.3	24.8	18.5	11.2	5.5
U2 - cutoff = 0.95	5	34.7	26.7	18.8	10.5	4.3
U3 - cutoff = 0.75	7.7	40.7	22.7	15.4	6.4	7.1
U3 - cutoff = 0.85	6.9	42.5	22.5	16.3	6.3	5.5
U3 - cutoff = 0.95	8.5	41.7	23.2	15.5	6.7	4.3
Mean administrations						
U2 - cutoff = 0.75	4.1	6.2	6.5	6.2	5.5	-
U2 - cutoff = 0.85	4.5	6.5	6.5	5.8	5.6	-
U2 - cutoff = 0.95	5	6.5	6.6	5.8	5.5	-
U3 - cutoff = 0.75	4.1	6.2	6.4	6.2	5.6	-
U3 - cutoff = 0.85	4.5	6.5	6.6	5.7	5.6	-
U3 - cutoff = 0.95	5	6.5	6.7	5.8	5.5	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.5	3.4	2	2.3	1.9	-
U2 - cutoff = 0.85	0.7	3.5	2.1	2.2	1.9	-
U2 - cutoff = 0.95	0.2	3.4	1.8	2.1	1.1	-
U3 - cutoff = 0.75	0.6	3.3	2	2.4	1.9	-
U3 - cutoff = 0.85	0.7	3.4	2.1	2.2	1.9	-
U3 - cutoff = 0.95	0.2	3.4	1.7	2.1	1.1	-
Mean trial duration (days)						
U2 - cutoff = 0.75						Days
U2 - cutoff = 0.85						1007.6
U2 - cutoff = 0.95						1021.1
U3 - cutoff = 0.75						1151.9
U3 - cutoff = 0.85						1007.4
U3 - cutoff = 0.95						1020.3
						1150.9
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - cutoff = 0.75	58.3	13.4	4	1.4	0.5	22.4
U2 - cutoff = 0.85	66.7	13	3.6	1.3	0.4	15
U2 - cutoff = 0.95	68.2	12.9	3.6	0.9	0.6	13.8
U3 - cutoff = 0.75	61	11.9	3.2	1	0.4	22.5
U3 - cutoff = 0.85	67.3	13.2	3.3	1	0.4	14.8
U3 - cutoff = 0.95	69	13	3.2	0.9	0.3	13.6
Mean administrations						
U2 - cutoff = 0.75	17.8	3.9	2.5	1.1	0.4	-
U2 - cutoff = 0.85	19.2	4	2.5	1.1	0.4	-
U2 - cutoff = 0.95	19.5	4	2.5	1.1	0.4	-
U3 - cutoff = 0.75	17.8	3.9	2.5	1.1	0.4	-
U3 - cutoff = 0.85	19.1	4	2.6	1.1	0.4	-
U3 - cutoff = 0.95	19.5	4	2.5	1.1	0.4	-
Mean IPDE administrations						
U2 - cutoff = 0.75	4.8	1.5	0.7	0.4	0.1	-
U2 - cutoff = 0.85	5.2	1.5	0.7	0.4	0.1	-
U2 - cutoff = 0.95	0.4	1.5	0.6	0.3	0.1	-
U3 - cutoff = 0.75	4.8	1.5	0.7	0.4	0.1	-
U3 - cutoff = 0.85	5.2	1.6	0.8	0.4	0.1	-
U3 - cutoff = 0.95	0.4	1.5	0.5	0.3	0.1	-
Mean trial duration (days)						
U2 - cutoff = 0.75						Days
U2 - cutoff = 0.85						1001.4
U2 - cutoff = 0.95						1056.7
U3 - cutoff = 0.75						1368.1
U3 - cutoff = 0.85						1001
U3 - cutoff = 0.95						1057.1
						1367.3

Phase I/II Operating Characteristics

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - cutoff = 0.75	17	35.3	20.3	10	4.3	13.1
U2 - cutoff = 0.85	22.1	37.1	20.3	11	3.6	5.9
U2 - cutoff = 0.95	26.2	36.7	20.3	10.5	4	2.3
U3 - cutoff = 0.75	18.8	38.6	18	8.7	2.8	13.1
U3 - cutoff = 0.85	25.1	39.8	18.1	8.5	2.7	5.8
U3 - cutoff = 0.95	27.7	41.3	17.3	8.9	2.5	2.4
Mean administrations						
U2 - cutoff = 0.75	8.9	7	5.5	3.7	2.5	-
U2 - cutoff = 0.85	10.1	7	5.5	3.8	2.5	-
U2 - cutoff = 0.95	10.8	7	5.6	3.7	2.4	-
U3 - cutoff = 0.75	8.9	6.9	5.6	3.7	2.6	-
U3 - cutoff = 0.85	10.1	7	5.6	3.8	2.5	-
U3 - cutoff = 0.95	10.8	7.1	5.6	3.7	2.5	-
Mean IPDE administrations						
U2 - cutoff = 0.75	2.3	2.8	1.3	1.1	0.7	-
U2 - cutoff = 0.85	2.8	2.8	1.4	1.1	0.7	-
U2 - cutoff = 0.95	0.2	2.6	1.1	1	0.4	-
U3 - cutoff = 0.75	2.3	2.8	1.4	1.1	0.7	-
U3 - cutoff = 0.85	2.8	2.8	1.4	1.1	0.7	-
U3 - cutoff = 0.95	0.2	2.6	1.1	0.9	0.4	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1075.3
U2 - cutoff = 0.85						1118.8
U2 - cutoff = 0.95						1349.9
U3 - cutoff = 0.75						1075.6
U3 - cutoff = 0.85						1119.8
U3 - cutoff = 0.95						1347.2
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - cutoff = 0.75	13.7	5.4	2.8	1.1	0.3	76.7
U2 - cutoff = 0.85	26.2	8.5	3.2	1	0.4	60.7
U2 - cutoff = 0.95	43.3	9.3	2	1.2	0.4	43.8
U3 - cutoff = 0.75	14.9	5.6	1.7	1	0.2	76.6
U3 - cutoff = 0.85	27.4	8.8	2.3	0.9	0.3	60.3
U3 - cutoff = 0.95	43.7	9.2	1.8	1.1	0.3	43.9
Mean administrations						
U2 - cutoff = 0.75	9.9	3.3	1.9	0.8	0.3	-
U2 - cutoff = 0.85	13.4	3.6	1.9	0.8	0.3	-
U2 - cutoff = 0.95	17.4	3.7	1.8	0.8	0.3	-
U3 - cutoff = 0.75	9.9	3.3	1.8	0.9	0.3	-
U3 - cutoff = 0.85	13.4	3.6	1.9	0.8	0.3	-
U3 - cutoff = 0.95	17.4	3.7	1.9	0.7	0.3	-
Mean IPDE administrations						
U2 - cutoff = 0.75	2.7	1.5	0.6	0.3	0.1	-
U2 - cutoff = 0.85	3.9	1.6	0.6	0.3	0.1	-
U2 - cutoff = 0.95	0.3	1.6	0.4	0.3	0	-
U3 - cutoff = 0.75	2.7	1.6	0.6	0.3	0.1	-
U3 - cutoff = 0.85	3.9	1.7	0.6	0.3	0.1	-
U3 - cutoff = 0.95	0.4	1.6	0.4	0.2	0.1	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						598.2
U2 - cutoff = 0.85						733.4
U2 - cutoff = 0.95						1173.8
U3 - cutoff = 0.75						598.1
U3 - cutoff = 0.85						732.9
U3 - cutoff = 0.95						1174.1

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - cutoff = 0.75	4.7	5.1	15.3	13.9	25.7	35.3
U2 - cutoff = 0.85	7.7	8.6	17.8	13.6	26.4	25.9
U2 - cutoff = 0.95	13.6	10.7	16.9	13.5	26	19.3
U3 - cutoff = 0.75	9.4	6.9	13.9	11	23.3	35.5
U3 - cutoff = 0.85	13.9	9.8	15.1	12.7	22.5	26
U3 - cutoff = 0.95	20.8	11.5	14.9	11.8	21.8	19.3
Mean administrations						
U2 - cutoff = 0.75	5.6	3.8	5.2	4	5.2	-
U2 - cutoff = 0.85	6.9	4.1	5.3	4.1	5.2	-
U2 - cutoff = 0.95	8.9	4.3	5.4	4	5.1	-
U3 - cutoff = 0.75	5.6	3.8	5.1	4	5.2	-
U3 - cutoff = 0.85	6.8	4.1	5.3	4.1	5.3	-
U3 - cutoff = 0.95	8.9	4.3	5.4	4	5.1	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.2	2.2	1.8	1.6	1.5	-
U2 - cutoff = 0.85	1.7	2.4	1.8	1.6	1.5	-
U2 - cutoff = 0.95	0.4	2.7	1.3	1.4	0.7	-
U3 - cutoff = 0.75	1.2	2.2	1.7	1.6	1.5	-
U3 - cutoff = 0.85	1.7	2.4	1.8	1.6	1.5	-
U3 - cutoff = 0.95	0.4	2.6	1.3	1.4	0.7	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						847.8
U2 - cutoff = 0.85						908.6
U2 - cutoff = 0.95						1169.3
U3 - cutoff = 0.75						846.7
U3 - cutoff = 0.85						908.7
U3 - cutoff = 0.95						1168.7
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - cutoff = 0.75	4.1	25.3	12	4	2.2	52.4
U2 - cutoff = 0.85	6.4	29.1	12	4.3	1.9	46.3
U2 - cutoff = 0.95	11.7	31.1	12.4	4.3	2	38.5
U3 - cutoff = 0.75	4.8	26.4	10.7	4	1.4	52.7
U3 - cutoff = 0.85	6.9	30.5	10.7	4	1.6	46.3
U3 - cutoff = 0.95	12.4	33.3	10.3	3.7	1.3	39
Mean administrations						
U2 - cutoff = 0.75	6.2	5.7	4	2.1	1.2	-
U2 - cutoff = 0.85	8.4	5.9	4.1	2.2	1.1	-
U2 - cutoff = 0.95	11.8	5.8	4.2	2.1	1.1	-
U3 - cutoff = 0.75	6.2	5.6	4	2.2	1.2	-
U3 - cutoff = 0.85	8.4	6	4.1	2.2	1.2	-
U3 - cutoff = 0.95	11.7	6.1	4	2	1.1	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.4	2.4	1.1	0.6	0.3	-
U2 - cutoff = 0.85	2.2	2.5	1.1	0.7	0.3	-
U2 - cutoff = 0.95	0.1	2.3	0.8	0.6	0.2	-
U3 - cutoff = 0.75	1.4	2.4	1.1	0.7	0.3	-
U3 - cutoff = 0.85	2.2	2.5	1.1	0.6	0.3	-
U3 - cutoff = 0.95	0.1	2.3	0.8	0.6	0.2	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						734.7
U2 - cutoff = 0.85						817.3
U2 - cutoff = 0.95						1165.2
U3 - cutoff = 0.75						733.7
U3 - cutoff = 0.85						817.3
U3 - cutoff = 0.95						1163.7

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - cutoff = 0.75	45.2	18.3	3.7	0.8	0	32
U2 - cutoff = 0.85	58.1	20.8	4.6	1	0.1	15.4
U2 - cutoff = 0.95	66.6	21.4	4.1	0.7	0	7.2
U3 - cutoff = 0.75	47.8	16.9	2.5	0.8	0	32
U3 - cutoff = 0.85	60.4	19.2	4.4	0.6	0	15.4
U3 - cutoff = 0.95	68.3	20.9	3.2	0.7	0	7
Mean administrations						
U2 - cutoff = 0.75	14.4	5.7	3.4	1	0.2	-
U2 - cutoff = 0.85	16.8	5.9	3.5	1.1	0.2	-
U2 - cutoff = 0.95	18.5	5.8	3.5	1	0.1	-
U3 - cutoff = 0.75	14.3	5.7	3.5	1.1	0.2	-
U3 - cutoff = 0.85	16.9	5.7	3.6	1	0.2	-
U3 - cutoff = 0.95	18.4	5.9	3.5	1	0.1	-
Mean IPDE administrations						
U2 - cutoff = 0.75	4.2	2.5	1.1	0.4	0	-
U2 - cutoff = 0.85	5.1	2.5	1.1	0.4	0.1	-
U2 - cutoff = 0.95	0.6	2.4	0.8	0.4	0	-
U3 - cutoff = 0.75	4.2	2.5	1.1	0.4	0.1	-
U3 - cutoff = 0.85	5.2	2.5	1.1	0.4	0.1	-
U3 - cutoff = 0.95	0.6	2.4	0.8	0.4	0	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						907.1
U2 - cutoff = 0.85						1003.3
U2 - cutoff = 0.95						1370.3
U3 - cutoff = 0.75						908.1
U3 - cutoff = 0.85						1002.3
U3 - cutoff = 0.95						1366.6
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - cutoff = 0.75	10.6	9.2	7.7	36.3	24	12.2
U2 - cutoff = 0.85	12.2	11.4	9.6	37.6	23.8	5.4
U2 - cutoff = 0.95	15.1	12.2	9.4	36.3	23.8	3.2
U3 - cutoff = 0.75	15.4	9.6	6.7	34	22.4	11.9
U3 - cutoff = 0.85	18.8	11.5	7.7	34.7	21.8	5.5
U3 - cutoff = 0.95	21.6	12.5	7	34	21.5	3.5
Mean administrations						
U2 - cutoff = 0.75	4.2	4.5	5	7.4	7.3	-
U2 - cutoff = 0.85	4.7	4.5	5.2	7.5	7.3	-
U2 - cutoff = 0.95	5.1	4.6	5.3	7.4	7.1	-
U3 - cutoff = 0.75	4.2	4.4	5	7.4	7.4	-
U3 - cutoff = 0.85	4.7	4.5	5.3	7.4	7.2	-
U3 - cutoff = 0.95	5.1	4.6	5.4	7.4	7	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.6	2.8	1.6	2.7	1.6	-
U2 - cutoff = 0.85	0.8	2.8	1.6	2.8	1.6	-
U2 - cutoff = 0.95	0.3	2.9	1.4	2.2	0.8	-
U3 - cutoff = 0.75	0.6	2.8	1.6	2.7	1.6	-
U3 - cutoff = 0.85	0.8	2.9	1.7	2.7	1.6	-
U3 - cutoff = 0.95	0.4	2.9	1.4	2.3	0.8	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1050.2
U2 - cutoff = 0.85						1074.5
U2 - cutoff = 0.95						1205.8
U3 - cutoff = 0.75						1051.3
U3 - cutoff = 0.85						1074
U3 - cutoff = 0.95						1207.3

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - cutoff = 0.75	6	25.1	13.6	4.7	2	48.6
U2 - cutoff = 0.85	8.3	29.4	13.8	5.1	2	41.4
U2 - cutoff = 0.95	13.4	33.3	13.3	4.3	1.8	33.9
U3 - cutoff = 0.75	7	26.8	12	4.3	1.4	48.5
U3 - cutoff = 0.85	9.5	30.2	12.9	4.6	1.3	41.5
U3 - cutoff = 0.95	15	33.1	12.2	4.1	1.5	34
Mean administrations						
U2 - cutoff = 0.75	6.2	5.9	4.5	2.4	1.3	-
U2 - cutoff = 0.85	8.2	6.2	4.4	2.4	1.2	-
U2 - cutoff = 0.95	11.5	6.2	4.5	2.4	1.2	-
U3 - cutoff = 0.75	6.2	5.9	4.4	2.4	1.2	-
U3 - cutoff = 0.85	8.2	6.2	4.5	2.4	1.2	-
U3 - cutoff = 0.95	11.3	6.2	4.5	2.4	1.2	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.4	2.7	1.3	0.8	0.4	-
U2 - cutoff = 0.85	2.2	2.8	1.3	0.8	0.4	-
U2 - cutoff = 0.95	0.1	2.5	1	0.7	0.2	-
U3 - cutoff = 0.75	1.4	2.7	1.3	0.8	0.4	-
U3 - cutoff = 0.85	2.1	2.8	1.4	0.8	0.4	-
U3 - cutoff = 0.95	0.1	2.5	1	0.7	0.2	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						747.9
U2 - cutoff = 0.85						822
U2 - cutoff = 0.95						1162.3
U3 - cutoff = 0.75						748
U3 - cutoff = 0.85						822.3
U3 - cutoff = 0.95						1161.2
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - cutoff = 0.75	12.4	8.7	10.2	28.8	16.3	23.6
U2 - cutoff = 0.85	16.2	11.7	13.7	29	16.6	12.8
U2 - cutoff = 0.95	20.7	14	13	29.6	15.3	7.4
U3 - cutoff = 0.75	18.3	12.1	8.4	24.5	12.9	23.8
U3 - cutoff = 0.85	24.8	14.3	9.8	26.3	11.9	12.9
U3 - cutoff = 0.95	31.8	15.3	9.1	24.7	11.5	7.6
Mean administrations						
U2 - cutoff = 0.75	5.1	4.7	5.4	6.2	5.1	-
U2 - cutoff = 0.85	5.8	5.1	5.7	6.2	5.2	-
U2 - cutoff = 0.95	6.8	5.3	5.8	6.1	5	-
U3 - cutoff = 0.75	5	4.7	5.5	6.3	5.1	-
U3 - cutoff = 0.85	5.9	5.1	5.7	6.2	5.2	-
U3 - cutoff = 0.95	6.8	5.4	5.8	6.1	4.9	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1	2.8	1.8	2.4	1.5	-
U2 - cutoff = 0.85	1.3	3	2	2.4	1.5	-
U2 - cutoff = 0.95	0.6	3.1	1.5	2	0.8	-
U3 - cutoff = 0.75	0.9	2.8	1.9	2.4	1.5	-
U3 - cutoff = 0.85	1.3	3	1.9	2.4	1.5	-
U3 - cutoff = 0.95	0.6	3.1	1.5	2.1	0.8	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						937
U2 - cutoff = 0.85						984.3
U2 - cutoff = 0.95						1164
U3 - cutoff = 0.75						938.6
U3 - cutoff = 0.85						984.3
U3 - cutoff = 0.95						1161.2

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - cutoff = 0.75	30.3	15.2	8.2	3.3	0.9	42.1
U2 - cutoff = 0.85	43.2	23.6	11.2	3.2	0.9	17.9
U2 - cutoff = 0.95	54.2	23.6	10.1	3.7	1.3	7.1
U3 - cutoff = 0.75	37.2	15	5	1.3	0.4	41.1
U3 - cutoff = 0.85	52.3	20.1	6.9	2.3	0.4	18
U3 - cutoff = 0.95	64.2	21	5.4	1.7	0.5	7.2
Mean administrations						
U2 - cutoff = 0.75	8.1	6	5.5	3.4	1.7	-
U2 - cutoff = 0.85	10	6.5	5.8	3.3	1.7	-
U2 - cutoff = 0.95	11.7	6.7	5.8	3.4	1.6	-
U3 - cutoff = 0.75	8.2	5.9	5.6	3.4	1.7	-
U3 - cutoff = 0.85	10	6.6	5.8	3.3	1.7	-
U3 - cutoff = 0.95	11.8	6.7	5.7	3.3	1.6	-
Mean IPDE administrations						
U2 - cutoff = 0.75	2.2	3.1	1.9	1.5	0.6	-
U2 - cutoff = 0.85	3	3.4	2	1.4	0.7	-
U2 - cutoff = 0.95	1	3.4	1.6	1.3	0.4	-
U3 - cutoff = 0.75	2.3	3.1	1.9	1.4	0.6	-
U3 - cutoff = 0.85	3	3.4	2	1.4	0.6	-
U3 - cutoff = 0.95	1	3.4	1.6	1.3	0.4	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						851
U2 - cutoff = 0.85						932.8
U2 - cutoff = 0.95						1181.7
U3 - cutoff = 0.75						852.8
U3 - cutoff = 0.85						934.1
U3 - cutoff = 0.95						1181.5
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - cutoff = 0.75	31.1	24.9	16.8	14.1	12.5	0.6
U2 - cutoff = 0.85	31.3	23	16.7	16	12.9	0.1
U2 - cutoff = 0.95	30.6	23.5	16	17.2	12.6	0.1
U3 - cutoff = 0.75	38.8	26.8	16.9	10.3	6.6	0.6
U3 - cutoff = 0.85	40.1	24.6	15.8	12.3	7.1	0.1
U3 - cutoff = 0.95	39.2	26.6	14.6	12.6	7	0.1
Mean administrations						
U2 - cutoff = 0.75	5.5	4.9	5.1	7.1	7.3	-
U2 - cutoff = 0.85	6.1	5.4	5.5	5.8	7.3	-
U2 - cutoff = 0.95	6	5.5	5.5	5.9	7.2	-
U3 - cutoff = 0.75	5.5	4.9	5.2	7.1	7.2	-
U3 - cutoff = 0.85	6	5.5	5.5	5.8	7.3	-
U3 - cutoff = 0.95	5.9	5.5	5.5	5.9	7.2	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1	2.9	1.7	3.2	2.8	-
U2 - cutoff = 0.85	1.3	3.1	1.9	2.6	2.8	-
U2 - cutoff = 0.95	1.1	3.1	1.8	2.6	1.5	-
U3 - cutoff = 0.75	1	2.9	1.8	3.2	2.7	-
U3 - cutoff = 0.85	1.3	3.1	1.9	2.6	2.8	-
U3 - cutoff = 0.95	1	3.1	1.8	2.7	1.5	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1010
U2 - cutoff = 0.85						1009
U2 - cutoff = 0.95						1095.5
U3 - cutoff = 0.75						1009.4
U3 - cutoff = 0.85						1008.4
U3 - cutoff = 0.95						1095.5

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - cutoff = 0.75	28.9	48.3	16.9	2.3	0	3.6
U2 - cutoff = 0.85	30	49.3	17.7	2.1	0	0.9
U2 - cutoff = 0.95	31.2	49.6	17	1.9	0	0.3
U3 - cutoff = 0.75	32	48.4	14.1	1.8	0.1	3.6
U3 - cutoff = 0.85	33.4	48.9	14.7	1.9	0.1	1
U3 - cutoff = 0.95	33.8	50.9	13.5	1.3	0.1	0.3
Mean administrations						
U2 - cutoff = 0.75	9.6	9.7	7.3	2.3	0.5	-
U2 - cutoff = 0.85	10.1	9.7	7.2	2.4	0.4	-
U2 - cutoff = 0.95	10.3	9.7	7.2	2.3	0.4	-
U3 - cutoff = 0.75	9.6	9.8	7.2	2.4	0.4	-
U3 - cutoff = 0.85	10.1	9.7	7.3	2.3	0.4	-
U3 - cutoff = 0.95	10.3	9.7	7.2	2.4	0.4	-
Mean IPDE administrations						
U2 - cutoff = 0.75	2.3	3.4	1.6	0.6	0.1	-
U2 - cutoff = 0.85	2.5	3.4	1.6	0.6	0.1	-
U2 - cutoff = 0.95	0.4	2.8	1.1	0.6	0.1	-
U3 - cutoff = 0.75	2.3	3.4	1.6	0.6	0.1	-
U3 - cutoff = 0.85	2.5	3.4	1.6	0.6	0.1	-
U3 - cutoff = 0.95	0.4	2.8	1.1	0.6	0.1	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1176.2
U2 - cutoff = 0.85						1191.4
U2 - cutoff = 0.95						1378.2
U3 - cutoff = 0.75						1175.8
U3 - cutoff = 0.85						1190.8
U3 - cutoff = 0.95						1373.9
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - cutoff = 0.75	9.8	8.4	9.4	13.8	56.3	2.3
U2 - cutoff = 0.85	9.1	8.3	9.6	13.6	58.3	1.1
U2 - cutoff = 0.95	11.6	6.5	9.5	13.8	57.9	0.7
U3 - cutoff = 0.75	15.5	10.7	9	11.9	50.8	2.1
U3 - cutoff = 0.85	15.4	11.5	10.5	11.2	50.4	1
U3 - cutoff = 0.95	16.1	11	9.2	13.2	49.7	0.8
Mean administrations						
U2 - cutoff = 0.75	3.6	3.8	4.6	5.6	12.1	-
U2 - cutoff = 0.85	3.7	3.9	4.6	5.5	12.2	-
U2 - cutoff = 0.95	3.7	3.9	4.7	5.7	11.9	-
U3 - cutoff = 0.75	3.6	3.8	4.6	5.5	12.2	-
U3 - cutoff = 0.85	3.7	3.8	4.6	5.6	12.1	-
U3 - cutoff = 0.95	3.8	3.9	4.6	5.6	12	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.3	2.7	1.5	2.5	3.3	-
U2 - cutoff = 0.85	0.3	2.7	1.5	2.5	3.4	-
U2 - cutoff = 0.95	0.2	2.8	1.4	2.3	1.4	-
U3 - cutoff = 0.75	0.3	2.7	1.5	2.5	3.4	-
U3 - cutoff = 0.85	0.3	2.7	1.5	2.5	3.4	-
U3 - cutoff = 0.95	0.2	2.8	1.4	2.3	1.4	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1069.1
U2 - cutoff = 0.85						1073.3
U2 - cutoff = 0.95						1199
U3 - cutoff = 0.75						1069.9
U3 - cutoff = 0.85						1071.8
U3 - cutoff = 0.95						1201.5

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - cutoff = 0.75	11.9	44.9	16	2.2	0.1	24.9
U2 - cutoff = 0.85	16.1	50.6	17	1.7	0	14.6
U2 - cutoff = 0.95	19.9	52.8	17.2	1.5	0.1	8.5
U3 - cutoff = 0.75	14.9	44.9	13.2	1.7	0.1	25.2
U3 - cutoff = 0.85	18.8	51	14.1	1.6	0	14.5
U3 - cutoff = 0.95	23.4	52.5	14	1.6	0.1	8.4
Mean administrations						
U2 - cutoff = 0.75	6.3	9.8	6.9	2.2	0.4	-
U2 - cutoff = 0.85	7.6	10.2	7	2.2	0.4	-
U2 - cutoff = 0.95	9.1	10.3	7	2.2	0.3	-
U3 - cutoff = 0.75	6.3	9.8	6.9	2.2	0.4	-
U3 - cutoff = 0.85	7.6	10.2	6.9	2.2	0.4	-
U3 - cutoff = 0.95	9.2	10.4	6.9	2.1	0.3	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.4	4.2	1.9	0.8	0.1	-
U2 - cutoff = 0.85	1.9	4.4	2	0.8	0.1	-
U2 - cutoff = 0.95	0.4	3.6	1.3	0.7	0.1	-
U3 - cutoff = 0.75	1.4	4.2	1.9	0.8	0.1	-
U3 - cutoff = 0.85	1.9	4.4	1.9	0.8	0.1	-
U3 - cutoff = 0.95	0.4	3.6	1.3	0.7	0.1	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						947.9
U2 - cutoff = 0.85						1006.2
U2 - cutoff = 0.95						1268.7
U3 - cutoff = 0.75						949
U3 - cutoff = 0.85						1005.2
U3 - cutoff = 0.95						1266.2
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - cutoff = 0.75	10.1	41.6	25.7	9.5	1.4	11.7
U2 - cutoff = 0.85	10.5	44	28.8	9.1	1.4	6.2
U2 - cutoff = 0.95	12.1	45.3	27.3	9.9	1.2	4.2
U3 - cutoff = 0.75	16.1	42.2	21.4	7.4	0.9	12
U3 - cutoff = 0.85	16.4	46.5	22.6	7.6	0.8	6.1
U3 - cutoff = 0.95	18.4	47.4	22.7	6.1	0.9	4.4
Mean administrations						
U2 - cutoff = 0.75	4.7	8.3	8.2	4.7	2.2	-
U2 - cutoff = 0.85	5.1	8.6	8.2	4.7	2.3	-
U2 - cutoff = 0.95	5.7	8.7	8.3	4.5	2.2	-
U3 - cutoff = 0.75	4.7	8.3	8.2	4.7	2.3	-
U3 - cutoff = 0.85	5.1	8.6	8.3	4.6	2.2	-
U3 - cutoff = 0.95	5.7	8.6	8.4	4.5	2.1	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.8	4.1	2.5	1.7	0.7	-
U2 - cutoff = 0.85	1	4.2	2.5	1.7	0.7	-
U2 - cutoff = 0.95	0.4	3.7	1.8	1.6	0.5	-
U3 - cutoff = 0.75	0.8	4.1	2.5	1.7	0.7	-
U3 - cutoff = 0.85	1	4.3	2.5	1.7	0.7	-
U3 - cutoff = 0.95	0.4	3.7	1.8	1.6	0.4	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						999.3
U2 - cutoff = 0.85						1025.2
U2 - cutoff = 0.95						1186.1
U3 - cutoff = 0.75						1000.8
U3 - cutoff = 0.85						1024.5
U3 - cutoff = 0.95						1185.6

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - cutoff = 0.75	60.9	1.5	0	0	0	37.6
U2 - cutoff = 0.85	61.8	1.6	0	0	0	36.6
U2 - cutoff = 0.95	60.1	1.4	0	0	0	38.5
U3 - cutoff = 0.75	61.4	1.4	0	0	0	37.2
U3 - cutoff = 0.85	62.1	1.3	0	0	0	36.6
U3 - cutoff = 0.95	59.7	1.5	0	0	0	38.8
Mean administrations						
U2 - cutoff = 0.75	20.9	2.1	0.6	0	0	-
U2 - cutoff = 0.85	21.2	2.1	0.6	0	0	-
U2 - cutoff = 0.95	20.8	2.1	0.6	0	0	-
U3 - cutoff = 0.75	20.9	2.1	0.6	0	0	-
U3 - cutoff = 0.85	21.1	2.1	0.6	0	0	-
U3 - cutoff = 0.95	20.8	2.1	0.6	0	0	-
Mean IPDE administrations						
U2 - cutoff = 0.75	4.2	0.6	0.1	0	0	-
U2 - cutoff = 0.85	4.3	0.6	0.1	0	0	-
U2 - cutoff = 0.95	0.1	0.7	0.1	0	0	-
U3 - cutoff = 0.75	4.2	0.6	0.1	0	0	-
U3 - cutoff = 0.85	4.3	0.6	0.1	0	0	-
U3 - cutoff = 0.95	0.1	0.7	0.1	0	0	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1030.4
U2 - cutoff = 0.85						1040.8
U2 - cutoff = 0.95						1255.4
U3 - cutoff = 0.75						1032
U3 - cutoff = 0.85						1040.2
U3 - cutoff = 0.95						1252.4
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - cutoff = 0.75	17.9	42.8	10.5	1.5	0	27.3
U2 - cutoff = 0.85	22.7	52.9	13	1.8	0	9.6
U2 - cutoff = 0.95	26.9	55.4	11.6	1.6	0	4.5
U3 - cutoff = 0.75	20.4	42.1	8.8	1.5	0	27.2
U3 - cutoff = 0.85	25.8	52.2	10.8	1.4	0	9.8
U3 - cutoff = 0.95	31.1	53.5	9.3	1.5	0	4.6
Mean administrations						
U2 - cutoff = 0.75	6.6	11.1	6.4	1.7	0.3	-
U2 - cutoff = 0.85	7.6	12.1	6.3	1.8	0.3	-
U2 - cutoff = 0.95	8.7	12.2	6.5	1.7	0.2	-
U3 - cutoff = 0.75	6.6	11.2	6.4	1.7	0.3	-
U3 - cutoff = 0.85	7.6	12.1	6.5	1.7	0.3	-
U3 - cutoff = 0.95	8.8	12.3	6.4	1.6	0.3	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.4	4.8	1.9	0.6	0.1	-
U2 - cutoff = 0.85	1.8	5.3	1.8	0.6	0.1	-
U2 - cutoff = 0.95	0.6	4	1.3	0.5	0	-
U3 - cutoff = 0.75	1.4	4.9	1.8	0.5	0.1	-
U3 - cutoff = 0.85	1.8	5.2	1.9	0.6	0.1	-
U3 - cutoff = 0.95	0.6	4	1.3	0.5	0	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						956.5
U2 - cutoff = 0.85						1021.9
U2 - cutoff = 0.95						1261.1
U3 - cutoff = 0.75						958.1
U3 - cutoff = 0.85						1021.8
U3 - cutoff = 0.95						1259.1

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - cutoff = 0.75	1.5	0.5	0	0	0	98
U2 - cutoff = 0.85	7.2	3.9	0.4	0	0	88.5
U2 - cutoff = 0.95	19.7	5.2	0.2	0	0	74.9
U3 - cutoff = 0.75	1.5	0.4	0	0	0	98.1
U3 - cutoff = 0.85	7.3	3.7	0.4	0	0	88.6
U3 - cutoff = 0.95	19.7	5.1	0.1	0	0	75.1
Mean administrations						
U2 - cutoff = 0.75	8	2.1	0.6	0	0	-
U2 - cutoff = 0.85	11.8	2.5	0.6	0	0	-
U2 - cutoff = 0.95	17.2	2.7	0.6	0	0	-
U3 - cutoff = 0.75	8	2.2	0.6	0	0	-
U3 - cutoff = 0.85	11.8	2.5	0.6	0	0	-
U3 - cutoff = 0.95	17.2	2.7	0.6	0	0	-
Mean IPDE administrations						
U2 - cutoff = 0.75	2	1.1	0.2	0	0	-
U2 - cutoff = 0.85	3.3	1.3	0.2	0	0	-
U2 - cutoff = 0.95	0.3	1.5	0.2	0	0	-
U3 - cutoff = 0.75	2	1.1	0.2	0	0	-
U3 - cutoff = 0.85	3.3	1.3	0.2	0	0	-
U3 - cutoff = 0.95	0.3	1.5	0.2	0	0	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						397.8
U2 - cutoff = 0.85						547.3
U2 - cutoff = 0.95						1020.6
U3 - cutoff = 0.75						398.8
U3 - cutoff = 0.85						547.8
U3 - cutoff = 0.95						1019
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - cutoff = 0.75	1.3	0.9	5.7	45.7	45.9	0.5
U2 - cutoff = 0.85	1.2	1.1	5.8	45.4	46.4	0.1
U2 - cutoff = 0.95	1	1.7	4.9	44.8	47.5	0.1
U3 - cutoff = 0.75	2.5	2.4	7.6	46	41	0.5
U3 - cutoff = 0.85	2.8	2.2	7.5	47.3	40.1	0.1
U3 - cutoff = 0.95	2.2	2.6	6.6	48.4	40.1	0.1
Mean administrations						
U2 - cutoff = 0.75	3.3	3.2	3.7	8.4	11.3	-
U2 - cutoff = 0.85	3.3	3.2	3.8	8.4	11.3	-
U2 - cutoff = 0.95	3.3	3.3	3.8	8.3	11.3	-
U3 - cutoff = 0.75	3.3	3.2	3.8	8.4	11.3	-
U3 - cutoff = 0.85	3.3	3.2	3.8	8.4	11.3	-
U3 - cutoff = 0.95	3.3	3.2	3.8	8.4	11.3	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.1	2.5	1.2	3.3	3.1	-
U2 - cutoff = 0.85	0.1	2.5	1.2	3.3	3.1	-
U2 - cutoff = 0.95	0.1	2.5	1.2	3.1	1.2	-
U3 - cutoff = 0.75	0.1	2.5	1.2	3.3	3.1	-
U3 - cutoff = 0.85	0.1	2.5	1.2	3.3	3.1	-
U3 - cutoff = 0.95	0.1	2.5	1.2	3.1	1.2	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1095
U2 - cutoff = 0.85						1096.3
U2 - cutoff = 0.95						1206
U3 - cutoff = 0.75						1096
U3 - cutoff = 0.85						1096
U3 - cutoff = 0.95						1207.5

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - cutoff = 0.75	28.9	25.3	32.5	5.5	0	7.8
U2 - cutoff = 0.85	31.9	26.8	33.5	5.3	0	2.5
U2 - cutoff = 0.95	35.5	28	31	4.9	0	0.6
U3 - cutoff = 0.75	33.6	26.3	28.4	4	0	7.7
U3 - cutoff = 0.85	37.2	27.6	28.7	4.2	0	2.3
U3 - cutoff = 0.95	40.3	27.7	27.8	3.5	0	0.7
Mean administrations						
U2 - cutoff = 0.75	8.5	7.8	9.1	2.8	0.5	-
U2 - cutoff = 0.85	9.1	8.1	9.2	2.8	0.5	-
U2 - cutoff = 0.95	9.4	8	9.3	2.7	0.5	-
U3 - cutoff = 0.75	8.5	7.9	9.1	2.8	0.5	-
U3 - cutoff = 0.85	9.1	8	9.2	2.7	0.5	-
U3 - cutoff = 0.95	9.5	7.9	9.3	2.7	0.5	-
Mean IPDE administrations						
U2 - cutoff = 0.75	2.2	3.5	2.5	0.9	0.1	-
U2 - cutoff = 0.85	2.4	3.6	2.5	0.8	0.1	-
U2 - cutoff = 0.95	0.7	3.1	1.6	0.8	0.1	-
U3 - cutoff = 0.75	2.2	3.5	2.5	0.9	0.1	-
U3 - cutoff = 0.85	2.4	3.6	2.5	0.9	0.1	-
U3 - cutoff = 0.95	0.7	3.1	1.6	0.8	0.1	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1088.2
U2 - cutoff = 0.85						1113.3
U2 - cutoff = 0.95						1305
U3 - cutoff = 0.75						1086.2
U3 - cutoff = 0.85						1114.3
U3 - cutoff = 0.95						1303.1
Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - cutoff = 0.75	14.6	20.4	26.9	6.5	0.1	31.5
U2 - cutoff = 0.85	19.2	24	30.4	6	0	20.4
U2 - cutoff = 0.95	28.8	28.3	27.4	5.2	0.1	10.2
U3 - cutoff = 0.75	18.4	21.9	23	5.4	0	31.3
U3 - cutoff = 0.85	23.1	26.4	24.4	6	0.1	20
U3 - cutoff = 0.95	32.1	28.7	23.9	4.7	0.1	10.4
Mean administrations						
U2 - cutoff = 0.75	7.2	6.3	7.8	2.9	0.5	-
U2 - cutoff = 0.85	8.9	6.5	8	2.8	0.5	-
U2 - cutoff = 0.95	10.9	6.7	7.9	2.8	0.5	-
U3 - cutoff = 0.75	7.2	6.2	7.9	2.9	0.5	-
U3 - cutoff = 0.85	9	6.6	7.9	2.9	0.5	-
U3 - cutoff = 0.95	11	6.6	7.9	2.9	0.5	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.8	3	2.2	0.9	0.1	-
U2 - cutoff = 0.85	2.5	3.2	2.3	0.9	0.1	-
U2 - cutoff = 0.95	0.5	3	1.3	0.9	0.1	-
U3 - cutoff = 0.75	1.8	3	2.3	0.9	0.1	-
U3 - cutoff = 0.85	2.5	3.2	2.3	0.9	0.1	-
U3 - cutoff = 0.95	0.5	3	1.3	0.9	0.1	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						915
U2 - cutoff = 0.85						984.5
U2 - cutoff = 0.95						1278.6
U3 - cutoff = 0.75						918.2
U3 - cutoff = 0.85						985.3
U3 - cutoff = 0.95						1277.7

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - cutoff = 0.75	13.2	18.4	41	9.6	0.7	17.1
U2 - cutoff = 0.85	15.9	20.3	40.9	10.7	0.7	11.5
U2 - cutoff = 0.95	22	21.7	41.8	8.1	0.9	5.5
U3 - cutoff = 0.75	16.6	19.6	37.5	8.4	0.4	17.5
U3 - cutoff = 0.85	20.2	22.2	36.7	8.9	0.7	11.3
U3 - cutoff = 0.95	25.6	22.2	38.7	7.8	0.3	5.5
Mean administrations						
U2 - cutoff = 0.75	6	6	9.6	4	1.4	-
U2 - cutoff = 0.85	7.1	6.3	9.4	4.1	1.4	-
U2 - cutoff = 0.95	8.2	6.4	9.4	4	1.3	-
U3 - cutoff = 0.75	6	6	9.5	4	1.4	-
U3 - cutoff = 0.85	7	6.3	9.5	4	1.4	-
U3 - cutoff = 0.95	8.3	6.3	9.5	3.9	1.3	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.3	3	2.7	1.2	0.3	-
U2 - cutoff = 0.85	1.7	3.2	2.7	1.2	0.3	-
U2 - cutoff = 0.95	0.4	3	1.6	1.1	0.2	-
U3 - cutoff = 0.75	1.3	3	2.7	1.2	0.4	-
U3 - cutoff = 0.85	1.8	3.1	2.7	1.2	0.3	-
U3 - cutoff = 0.95	0.4	3	1.6	1.1	0.2	-
Mean trial duration (days)						
U2 - cutoff = 0.75						Days
U2 - cutoff = 0.85						1013.5
U2 - cutoff = 0.95						1050.3
U3 - cutoff = 0.75						1271.8
U3 - cutoff = 0.85						1013.4
U3 - cutoff = 0.95						1050.7
						1272.2
Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - cutoff = 0.75	8.6	29.7	5.3	0.5	0	55.9
U2 - cutoff = 0.85	14.8	35.1	6.6	0.2	0	43.3
U2 - cutoff = 0.95	21.3	38	6.5	0.2	0	34
U3 - cutoff = 0.75	10.3	28.9	4.6	0.4	0	55.8
U3 - cutoff = 0.85	15.5	35.5	5.5	0.3	0	43.2
U3 - cutoff = 0.95	21.7	39.4	5	0.3	0	33.6
Mean administrations						
U2 - cutoff = 0.75	6.7	7.9	4.1	0.9	0.1	-
U2 - cutoff = 0.85	8.7	8.4	4.3	0.9	0.1	-
U2 - cutoff = 0.95	12.1	8.4	4.2	0.9	0.1	-
U3 - cutoff = 0.75	6.6	7.9	4.2	0.9	0.1	-
U3 - cutoff = 0.85	8.8	8.4	4.2	0.9	0.1	-
U3 - cutoff = 0.95	12.1	8.5	4.1	0.9	0.1	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.5	3.2	1.2	0.3	0	-
U2 - cutoff = 0.85	2.3	3.4	1.2	0.3	0	-
U2 - cutoff = 0.95	0.3	2.7	0.9	0.3	0	-
U3 - cutoff = 0.75	1.5	3.2	1.2	0.3	0	-
U3 - cutoff = 0.85	2.3	3.4	1.2	0.3	0	-
U3 - cutoff = 0.95	0.3	2.7	0.8	0.3	0	-
Mean trial duration (days)						
U2 - cutoff = 0.75						Days
U2 - cutoff = 0.85						740.6
U2 - cutoff = 0.95						830.3
U3 - cutoff = 0.75						1194.1
U3 - cutoff = 0.85						742.6
U3 - cutoff = 0.95						829.2
						1193.2

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - cutoff = 0.75	65.4	25.1	4.3	0.3	0	4.9
U2 - cutoff = 0.85	67.2	25.6	4.2	0.3	0	2.7
U2 - cutoff = 0.95	68.6	24.5	4	0.3	0	2.6
U3 - cutoff = 0.75	68.1	23.5	3.1	0.2	0	5.1
U3 - cutoff = 0.85	69.8	23.7	3.4	0.3	0	2.8
U3 - cutoff = 0.95	71.3	22	3.7	0.3	0	2.8
Mean administrations						
U2 - cutoff = 0.75	17.7	7.3	3.5	0.6	0	-
U2 - cutoff = 0.85	18	7.3	3.6	0.5	0	-
U2 - cutoff = 0.95	18.2	7.2	3.5	0.5	0	-
U3 - cutoff = 0.75	17.7	7.2	3.6	0.5	0	-
U3 - cutoff = 0.85	18	7.3	3.6	0.5	0.1	-
U3 - cutoff = 0.95	18.1	7.2	3.6	0.5	0	-
Mean IPDE administrations						
U2 - cutoff = 0.75	4.5	2.4	0.9	0.2	0	-
U2 - cutoff = 0.85	4.6	2.4	1	0.2	0	-
U2 - cutoff = 0.95	0.6	2.1	0.8	0.2	0	-
U3 - cutoff = 0.75	4.6	2.3	1	0.2	0	-
U3 - cutoff = 0.85	4.6	2.4	1	0.2	0	-
U3 - cutoff = 0.95	0.6	2.1	0.8	0.2	0	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1162.4
U2 - cutoff = 0.85						1177.4
U2 - cutoff = 0.95						1435.4
U3 - cutoff = 0.75						1162.5
U3 - cutoff = 0.85						1178
U3 - cutoff = 0.95						1430.7
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - cutoff = 0.75	18.2	12.6	14.3	37.1	12.6	5.2
U2 - cutoff = 0.85	19.2	13.8	15.1	37.1	13.1	1.7
U2 - cutoff = 0.95	21.1	13.8	15.5	36.4	12.4	0.8
U3 - cutoff = 0.75	24	13.8	12.1	33.8	11.1	5.2
U3 - cutoff = 0.85	26.4	14.1	13.7	33.6	10.5	1.7
U3 - cutoff = 0.95	28.2	15.1	12.6	33	10.2	0.8
Mean administrations						
U2 - cutoff = 0.75	4.6	4.5	5.7	8.1	6.4	-
U2 - cutoff = 0.85	5	4.5	5.8	8	6.4	-
U2 - cutoff = 0.95	5.2	4.6	5.8	8.2	6.2	-
U3 - cutoff = 0.75	4.6	4.5	5.7	8.2	6.3	-
U3 - cutoff = 0.85	5	4.5	5.8	8	6.4	-
U3 - cutoff = 0.95	5.2	4.6	5.8	8.1	6.2	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.7	2.7	1.8	2.8	1.6	-
U2 - cutoff = 0.85	0.9	2.7	1.8	2.8	1.6	-
U2 - cutoff = 0.95	0.4	2.8	1.5	2.3	0.9	-
U3 - cutoff = 0.75	0.7	2.7	1.8	2.8	1.5	-
U3 - cutoff = 0.85	0.9	2.7	1.8	2.8	1.6	-
U3 - cutoff = 0.95	0.4	2.8	1.6	2.3	0.9	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1085.9
U2 - cutoff = 0.85						1097.5
U2 - cutoff = 0.95						1216.2
U3 - cutoff = 0.75						1084.3
U3 - cutoff = 0.85						1097.5
U3 - cutoff = 0.95						1215.3

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - cutoff = 0.75	7.6	33.6	25.6	25.4	7.2	0.6
U2 - cutoff = 0.85	7.2	34.9	24.8	24.9	8	0.2
U2 - cutoff = 0.95	7	36.1	26	24.3	6.5	0.1
U3 - cutoff = 0.75	9.8	41.4	23.4	20.6	4.2	0.6
U3 - cutoff = 0.85	10.2	41.6	23.7	20.5	3.8	0.2
U3 - cutoff = 0.95	9.8	42.2	23.3	20.8	3.8	0.1
Mean administrations						
U2 - cutoff = 0.75	4	5.7	6.4	7.7	6	-
U2 - cutoff = 0.85	4.1	5.7	6.4	7.7	6	-
U2 - cutoff = 0.95	4.1	5.9	6.5	7.6	5.9	-
U3 - cutoff = 0.75	4	5.7	6.4	7.7	6	-
U3 - cutoff = 0.85	4.1	5.7	6.4	7.7	6	-
U3 - cutoff = 0.95	4.1	5.7	6.5	7.7	5.9	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.4	2.8	1.6	2	1.5	-
U2 - cutoff = 0.85	0.5	2.8	1.6	2	1.5	-
U2 - cutoff = 0.95	0.1	2.8	1.4	1.7	1.1	-
U3 - cutoff = 0.75	0.4	2.8	1.6	2	1.5	-
U3 - cutoff = 0.85	0.5	2.8	1.6	2	1.5	-
U3 - cutoff = 0.95	0.2	2.8	1.5	1.7	1.1	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1188.1
U2 - cutoff = 0.85						1189.8
U2 - cutoff = 0.95						1256.2
U3 - cutoff = 0.75						1187.7
U3 - cutoff = 0.85						1188.8
U3 - cutoff = 0.95						1255
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - cutoff = 0.75	8.8	14.1	34.1	27.2	14.4	1.4
U2 - cutoff = 0.85	8.7	13.4	34.7	27.2	15.1	0.9
U2 - cutoff = 0.95	9.5	13.6	33.8	28	14.5	0.6
U3 - cutoff = 0.75	15.3	17.6	33.8	22.6	9.2	1.5
U3 - cutoff = 0.85	14.6	17.8	34.6	23.2	9	0.8
U3 - cutoff = 0.95	15.3	17.1	32.9	24.9	9.1	0.6
Mean administrations						
U2 - cutoff = 0.75	3.8	4.2	7.4	7.5	6.9	-
U2 - cutoff = 0.85	3.9	4.3	7.7	7.1	6.9	-
U2 - cutoff = 0.95	4	4.3	7.7	7.2	6.7	-
U3 - cutoff = 0.75	3.8	4.2	7.4	7.6	6.8	-
U3 - cutoff = 0.85	4	4.3	7.7	7	6.9	-
U3 - cutoff = 0.95	4	4.3	7.7	7.1	6.8	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.3	2.8	2.3	2.9	2.3	-
U2 - cutoff = 0.85	0.4	2.8	2.5	2.7	2.3	-
U2 - cutoff = 0.95	0.3	2.9	2	2.6	1.1	-
U3 - cutoff = 0.75	0.3	2.8	2.4	2.9	2.3	-
U3 - cutoff = 0.85	0.4	2.8	2.5	2.7	2.3	-
U3 - cutoff = 0.95	0.3	2.8	2.1	2.6	1.2	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1055
U2 - cutoff = 0.85						1054.9
U2 - cutoff = 0.95						1153.5
U3 - cutoff = 0.75						1055.2
U3 - cutoff = 0.85						1055
U3 - cutoff = 0.95						1153.7

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - cutoff = 0.75	8.8	4.6	12.8	39.7	33	1.1
U2 - cutoff = 0.85	8.6	5.1	11.6	41.1	33.3	0.3
U2 - cutoff = 0.95	9.1	4.5	12.8	40.5	33.1	0
U3 - cutoff = 0.75	11.9	7	13.5	40.5	25.9	1.2
U3 - cutoff = 0.85	12.3	6.9	13.6	40	27	0.2
U3 - cutoff = 0.95	13.6	6.1	12.9	40.1	27.2	0.1
Mean administrations						
U2 - cutoff = 0.75	3.4	3.6	5.1	8.2	9.6	-
U2 - cutoff = 0.85	3.5	3.6	5.2	8.1	9.6	-
U2 - cutoff = 0.95	3.5	3.6	5.3	8.1	9.4	-
U3 - cutoff = 0.75	3.4	3.6	5	8.2	9.6	-
U3 - cutoff = 0.85	3.5	3.6	5.2	8.1	9.6	-
U3 - cutoff = 0.95	3.5	3.6	5.3	8.2	9.3	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.2	2.6	1.6	3.1	2.4	-
U2 - cutoff = 0.85	0.2	2.6	1.6	3.1	2.4	-
U2 - cutoff = 0.95	0.2	2.6	1.4	2.8	1	-
U3 - cutoff = 0.75	0.2	2.6	1.5	3.1	2.4	-
U3 - cutoff = 0.85	0.2	2.6	1.6	3.1	2.4	-
U3 - cutoff = 0.95	0.2	2.6	1.4	2.8	1	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1105.2
U2 - cutoff = 0.85						1107.4
U2 - cutoff = 0.95						1209.9
U3 - cutoff = 0.75						1106
U3 - cutoff = 0.85						1106.9
U3 - cutoff = 0.95						1211.4
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - cutoff = 0.75	11.1	9.7	42.9	18.7	5.7	11.9
U2 - cutoff = 0.85	11.9	11.7	47	19.8	5.4	4.2
U2 - cutoff = 0.95	14.7	12.1	47.6	18.8	4.8	2
U3 - cutoff = 0.75	15.7	12.7	38.8	17.4	3.9	11.5
U3 - cutoff = 0.85	18.2	13.3	44.3	16.1	3.6	4.5
U3 - cutoff = 0.95	21.9	14.4	42.7	15.8	3.1	2
Mean administrations						
U2 - cutoff = 0.75	4.5	5	9.8	5.4	3.6	-
U2 - cutoff = 0.85	5	5.5	9.9	5.4	3.4	-
U2 - cutoff = 0.95	5.3	5.8	10	5.5	3.2	-
U3 - cutoff = 0.75	4.6	5	9.8	5.5	3.5	-
U3 - cutoff = 0.85	5	5.5	9.8	5.4	3.5	-
U3 - cutoff = 0.95	5.4	5.8	9.8	5.5	3.2	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.7	3	3.1	1.9	1.1	-
U2 - cutoff = 0.85	0.9	3.2	3.1	1.9	1.1	-
U2 - cutoff = 0.95	0.5	3.2	2	1.7	0.6	-
U3 - cutoff = 0.75	0.7	2.9	3.1	1.9	1.1	-
U3 - cutoff = 0.85	0.9	3.2	3.1	1.9	1.1	-
U3 - cutoff = 0.95	0.5	3.3	2	1.7	0.5	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1023.4
U2 - cutoff = 0.85						1049.8
U2 - cutoff = 0.95						1202.2
U3 - cutoff = 0.75						1025.4
U3 - cutoff = 0.85						1050.6
U3 - cutoff = 0.95						1197.8

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - cutoff = 0.75	10.1	39.5	32.2	13.5	4	0.7
U2 - cutoff = 0.85	11.6	40.2	31.4	12.3	4.3	0.2
U2 - cutoff = 0.95	11.7	40.5	31.4	11.8	4.5	0.1
U3 - cutoff = 0.75	15.9	43	28.3	9.4	2.7	0.7
U3 - cutoff = 0.85	16.1	44	27.2	9.9	2.6	0.2
U3 - cutoff = 0.95	16.8	44.5	26.7	9.6	2.4	0.1
Mean administrations						
U2 - cutoff = 0.75	4.5	8.2	8.9	5.1	3.2	-
U2 - cutoff = 0.85	4.6	8.3	8.8	5	3.3	-
U2 - cutoff = 0.95	4.6	8.4	8.9	5.1	3	-
U3 - cutoff = 0.75	4.5	8.2	8.9	5	3.2	-
U3 - cutoff = 0.85	4.6	8.2	8.9	5	3.2	-
U3 - cutoff = 0.95	4.7	8.4	8.8	5	3.1	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.6	3.6	2.2	1.5	0.8	-
U2 - cutoff = 0.85	0.7	3.7	2.2	1.5	0.9	-
U2 - cutoff = 0.95	0.3	3.1	1.6	1.4	0.5	-
U3 - cutoff = 0.75	0.6	3.7	2.2	1.5	0.9	-
U3 - cutoff = 0.85	0.6	3.6	2.2	1.5	0.8	-
U3 - cutoff = 0.95	0.3	3.1	1.6	1.3	0.5	-
Mean trial duration (days)						
U2 - cutoff = 0.75						Days
U2 - cutoff = 0.85						1164.7
U2 - cutoff = 0.95						1166.3
U3 - cutoff = 0.75						1271.9
U3 - cutoff = 0.85						1163.6
U3 - cutoff = 0.95						1166.4
						1273.3
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - cutoff = 0.75	10.5	27	14.7	6.8	1.9	39.1
U2 - cutoff = 0.85	14.6	36.6	16.1	7	1.8	23.9
U2 - cutoff = 0.95	20.9	41.6	14.4	6	1.7	15.4
U3 - cutoff = 0.75	14.1	28.1	11.7	5.7	1.1	39.3
U3 - cutoff = 0.85	19.1	37.8	12.6	5.9	1	23.6
U3 - cutoff = 0.95	27	39.8	11.5	5.3	1.2	15.2
Mean administrations						
U2 - cutoff = 0.75	6.2	7.2	5.6	2.8	1.4	-
U2 - cutoff = 0.85	7.8	8	5.7	2.9	1.4	-
U2 - cutoff = 0.95	9.7	8.3	5.8	2.8	1.3	-
U3 - cutoff = 0.75	6.2	7.2	5.6	2.8	1.4	-
U3 - cutoff = 0.85	7.6	8.2	5.7	2.8	1.4	-
U3 - cutoff = 0.95	9.6	8.4	5.8	2.8	1.3	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.3	3.5	1.8	1.1	0.4	-
U2 - cutoff = 0.85	2	3.8	1.8	1.1	0.4	-
U2 - cutoff = 0.95	0.4	3.6	1.2	1	0.2	-
U3 - cutoff = 0.75	1.3	3.5	1.8	1	0.4	-
U3 - cutoff = 0.85	1.9	3.9	1.8	1	0.4	-
U3 - cutoff = 0.95	0.4	3.5	1.2	0.9	0.2	-
Mean trial duration (days)						
U2 - cutoff = 0.75						Days
U2 - cutoff = 0.85						838
U2 - cutoff = 0.95						914.5
U3 - cutoff = 0.75						1187.4
U3 - cutoff = 0.85						838.3
U3 - cutoff = 0.95						918.1
						1188.5

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - cutoff = 0.75	14.3	32.9	11.6	3	0.5	37.7
U2 - cutoff = 0.85	20.4	45.6	14.7	2.7	0.5	16.1
U2 - cutoff = 0.95	27.2	48.4	13.1	3.4	0.7	7.2
U3 - cutoff = 0.75	19.4	32	8.9	1.9	0.3	37.5
U3 - cutoff = 0.85	25.5	44.6	11.2	2	0.4	16.3
U3 - cutoff = 0.95	33.2	47	9.8	2.5	0.3	7.2
Mean administrations						
U2 - cutoff = 0.75	6	9.2	6	2.4	0.9	-
U2 - cutoff = 0.85	7.3	10.3	6.2	2.4	0.9	-
U2 - cutoff = 0.95	8.7	10.7	6.2	2.3	0.8	-
U3 - cutoff = 0.75	6	9.2	6.1	2.4	0.9	-
U3 - cutoff = 0.85	7.2	10.3	6.2	2.4	0.9	-
U3 - cutoff = 0.95	8.8	10.6	6.2	2.4	0.8	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.2	4.4	1.9	1	0.3	-
U2 - cutoff = 0.85	1.7	4.8	1.9	1	0.3	-
U2 - cutoff = 0.95	0.6	4.2	1.3	0.9	0.2	-
U3 - cutoff = 0.75	1.2	4.4	1.9	1	0.3	-
U3 - cutoff = 0.85	1.7	4.8	1.9	0.9	0.3	-
U3 - cutoff = 0.95	0.6	4.1	1.3	0.9	0.2	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						875.9
U2 - cutoff = 0.85						955.2
U2 - cutoff = 0.95						1199.1
U3 - cutoff = 0.75						877.1
U3 - cutoff = 0.85						954.7
U3 - cutoff = 0.95						1198.3

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - cutoff = 0.75	8.6	3	2.4	0.5	0.8	84.7
U2 - cutoff = 0.85	16.4	11.5	2.3	0.6	0.2	69
U2 - cutoff = 0.95	35.3	16.1	2.4	0.6	0.1	45.5
U3 - cutoff = 0.75	7.9	3.6	3.1	0.6	0.7	84.1
U3 - cutoff = 0.85	16.7	11.4	1.8	0.4	0.1	69.6
U3 - cutoff = 0.95	34.3	15.7	1.6	0	0.1	48.3
Mean administrations						
U2 - cutoff = 0.75	7.5	4.8	2.2	1.3	0.4	-
U2 - cutoff = 0.85	10.7	5.7	2.5	1.4	0.4	-
U2 - cutoff = 0.95	14.7	6.2	2.4	1.4	0.4	-
U3 - cutoff = 0.75	7.6	4.8	2.2	1.3	0.4	-
U3 - cutoff = 0.85	10.8	5.6	2.5	1.4	0.4	-
U3 - cutoff = 0.95	15.1	5.8	2.3	1.3	0.4	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.5	2.6	0.7	0.6	0.2	-
U2 - cutoff = 0.85	2.8	2.9	0.8	0.6	0.2	-
U2 - cutoff = 0.95	1	3.4	0.7	0.6	0.1	-
U3 - cutoff = 0.75	1.6	2.5	0.7	0.6	0.1	-
U3 - cutoff = 0.85	2.9	2.9	0.8	0.6	0.2	-
U3 - cutoff = 0.95	1	3.2	0.7	0.6	0.1	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						586.6
U2 - cutoff = 0.85						732.2
U2 - cutoff = 0.95						1057.6
U3 - cutoff = 0.75						584.7
U3 - cutoff = 0.85						731.2
U3 - cutoff = 0.95						1064.2
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - cutoff = 0.75	20	16.3	37.7	14.6	2.2	9.2
U2 - cutoff = 0.85	23	17.3	39.6	15.4	2.1	2.6
U2 - cutoff = 0.95	25	19	39.3	13.4	2.2	1.1
U3 - cutoff = 0.75	24.7	17.3	34.6	12.1	1.9	9.4
U3 - cutoff = 0.85	28.1	19	35.2	13.1	1.7	2.9
U3 - cutoff = 0.95	30.4	20.8	35	11.3	1.5	1
Mean administrations						
U2 - cutoff = 0.75	8.2	6.7	8.7	3.9	0.9	-
U2 - cutoff = 0.85	9	6.9	8.9	3.9	0.9	-
U2 - cutoff = 0.95	9.2	7.1	8.9	3.8	0.9	-
U3 - cutoff = 0.75	8.7	6.8	8.6	3.6	0.7	-
U3 - cutoff = 0.85	9.3	7.1	8.8	3.6	0.7	-
U3 - cutoff = 0.95	9.9	7.3	8.6	3.4	0.7	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.9	3.1	2.4	1.1	0.3	-
U2 - cutoff = 0.85	2.2	3.2	2.4	1.1	0.3	-
U2 - cutoff = 0.95	0.5	3.2	1.7	0.9	0.2	-
U3 - cutoff = 0.75	2.1	3.2	2.3	1.1	0.2	-
U3 - cutoff = 0.85	2.3	3.3	2.4	1.1	0.2	-
U3 - cutoff = 0.95	0.6	3.2	1.6	0.9	0.2	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1086.2
U2 - cutoff = 0.85						1125.4
U2 - cutoff = 0.95						1288.9
U3 - cutoff = 0.75						1081.7
U3 - cutoff = 0.85						1120.4
U3 - cutoff = 0.95						1298.3

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - cutoff = 0.75	6.3	36.8	24.7	13.1	2.3	16.8
U2 - cutoff = 0.85	7.9	44.5	24.9	13.2	2.4	7.1
U2 - cutoff = 0.95	9.6	49.3	25.1	11.7	1.8	2.5
U3 - cutoff = 0.75	7.7	42.4	22.3	9	1.7	16.9
U3 - cutoff = 0.85	9.6	49.1	23.1	9.2	1.7	7.3
U3 - cutoff = 0.95	11.9	53.5	21.4	8.8	1.7	2.8
Mean administrations						
U2 - cutoff = 0.75	5.4	10.1	6.5	3.8	1.1	-
U2 - cutoff = 0.85	6.1	11.1	6.8	3.8	1.1	-
U2 - cutoff = 0.95	6.7	11.4	6.8	3.7	1	-
U3 - cutoff = 0.75	5.4	10.9	6.4	3.4	0.9	-
U3 - cutoff = 0.85	6.3	11.8	6.5	3.4	0.9	-
U3 - cutoff = 0.95	6.9	12.3	6.3	3.3	0.8	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.7	4.2	1.9	1.3	0.3	-
U2 - cutoff = 0.85	1.1	4.6	2	1.2	0.3	-
U2 - cutoff = 0.95	0.4	3.8	1.8	1.1	0.3	-
U3 - cutoff = 0.75	0.8	4.6	1.8	1.1	0.3	-
U3 - cutoff = 0.85	1.1	4.8	1.9	1.1	0.3	-
U3 - cutoff = 0.95	0.4	3.8	1.6	1	0.2	-
Mean trial duration (days)						
						Days
U2 - cutoff = 0.75						1018.8
U2 - cutoff = 0.85						1085
U2 - cutoff = 0.95						1237.3
U3 - cutoff = 0.75						1015.6
U3 - cutoff = 0.85						1081.2
U3 - cutoff = 0.95						1254
Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - cutoff = 0.75	5.7	4.1	16.3	52.8	16.9	4.2
U2 - cutoff = 0.85	5.7	5.2	16.9	53.3	17.6	1.3
U2 - cutoff = 0.95	6.7	6.7	17.1	53.3	16	0.2
U3 - cutoff = 0.75	8.2	6.4	16.1	51.3	13.4	4.6
U3 - cutoff = 0.85	8.6	7.7	16.4	51.3	14.9	1.1
U3 - cutoff = 0.95	10.3	8.1	16	52.5	13	0.2
Mean administrations						
U2 - cutoff = 0.75	3.8	4.3	6.5	10.6	3.9	-
U2 - cutoff = 0.85	4.1	4.5	6.7	10.6	3.9	-
U2 - cutoff = 0.95	4.2	4.7	6.7	10.7	3.7	-
U3 - cutoff = 0.75	3.9	4.4	6.6	10.6	3.5	-
U3 - cutoff = 0.85	4.3	4.6	6.7	10.7	3.6	-
U3 - cutoff = 0.95	4.5	4.6	6.8	10.7	3.4	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.3	2.9	1.9	3.3	1	-
U2 - cutoff = 0.85	0.5	3	1.9	3.3	1	-
U2 - cutoff = 0.95	0.2	3	1.6	2.4	0.8	-
U3 - cutoff = 0.75	0.3	2.9	1.9	3.4	0.9	-
U3 - cutoff = 0.85	0.5	3	1.9	3.4	0.9	-
U3 - cutoff = 0.95	0.2	3	1.5	2.4	0.7	-
Mean trial duration (days)						
						Days
U2 - cutoff = 0.75						1083.2
U2 - cutoff = 0.85						1107.3
U2 - cutoff = 0.95						1215.5
U3 - cutoff = 0.75						1077.6
U3 - cutoff = 0.85						1104.7
U3 - cutoff = 0.95						1224.7

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - cutoff = 0.75	5.6	35.9	25	18.8	11.2	3.5
U2 - cutoff = 0.85	5.4	37.6	25.7	19.5	10.4	1.4
U2 - cutoff = 0.95	5.4	39.7	24.2	19.8	10.6	0.3
U3 - cutoff = 0.75	9	42.3	21.6	17.7	6.1	3.3
U3 - cutoff = 0.85	9	42.6	23.2	17.5	6.2	1.5
U3 - cutoff = 0.95	8.3	45.5	22.9	17.1	5.9	0.3
Mean administrations						
U2 - cutoff = 0.75	3.6	7	6.9	7.1	4.6	-
U2 - cutoff = 0.85	3.8	7.2	7	7.1	4.6	-
U2 - cutoff = 0.95	3.9	7.4	7.1	7.1	4.5	-
U3 - cutoff = 0.75	3.7	7.7	7.1	6.8	4	-
U3 - cutoff = 0.85	3.9	7.9	7.2	6.8	4	-
U3 - cutoff = 0.95	4	8.2	7.1	6.8	3.8	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.2	3.7	2.2	2.7	1.7	-
U2 - cutoff = 0.85	0.3	3.8	2.3	2.7	1.6	-
U2 - cutoff = 0.95	0.1	3.1	1.9	2.3	1.1	-
U3 - cutoff = 0.75	0.2	4	2.3	2.6	1.4	-
U3 - cutoff = 0.85	0.3	4.1	2.3	2.6	1.4	-
U3 - cutoff = 0.95	0.2	3.2	1.9	2.1	0.9	-
Mean trial duration (days)						
U2 - cutoff = 0.75						Days
U2 - cutoff = 0.85						1035.1
U2 - cutoff = 0.95						1052.3
U3 - cutoff = 0.75						1179.3
U3 - cutoff = 0.85						1034.1
U3 - cutoff = 0.95						1049.9
						1195.7
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - cutoff = 0.75	55.8	18.2	4.7	1.9	0.5	18.9
U2 - cutoff = 0.85	61.9	17.6	5.1	1.9	0.5	13
U2 - cutoff = 0.95	63.7	17.2	4.3	2	0.5	12.3
U3 - cutoff = 0.75	57.3	17.9	3.8	1.5	0.4	19.1
U3 - cutoff = 0.85	62.7	17.8	4.2	1.5	0.4	13.4
U3 - cutoff = 0.95	65.5	16.8	3.5	1.2	0.4	12.6
Mean administrations						
U2 - cutoff = 0.75	16.2	6.2	2.7	1.4	0.3	-
U2 - cutoff = 0.85	17.1	6.3	2.7	1.3	0.3	-
U2 - cutoff = 0.95	17.5	6.1	2.7	1.3	0.3	-
U3 - cutoff = 0.75	16.7	6.1	2.5	1.2	0.3	-
U3 - cutoff = 0.85	17.6	6.2	2.4	1.2	0.3	-
U3 - cutoff = 0.95	18	6	2.4	1.2	0.2	-
Mean IPDE administrations						
U2 - cutoff = 0.75	4	2.5	0.8	0.4	0.1	-
U2 - cutoff = 0.85	4.3	2.5	0.8	0.4	0.1	-
U2 - cutoff = 0.95	0.7	2.6	0.7	0.4	0.1	-
U3 - cutoff = 0.75	4.2	2.4	0.7	0.4	0.1	-
U3 - cutoff = 0.85	4.5	2.5	0.7	0.4	0.1	-
U3 - cutoff = 0.95	0.7	2.5	0.6	0.4	0.1	-
Mean trial duration (days)						
U2 - cutoff = 0.75						Days
U2 - cutoff = 0.85						1045.2
U2 - cutoff = 0.95						1081.9
U3 - cutoff = 0.75						1296.1
U3 - cutoff = 0.85						1041.9
U3 - cutoff = 0.95						1078.3
						1302.1

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - cutoff = 0.75	9.5	44.5	22	12	3.8	8.2
U2 - cutoff = 0.85	11.8	46.8	22.7	12.5	3.5	2.7
U2 - cutoff = 0.95	12.2	48.3	23.7	11.6	3.2	1
U3 - cutoff = 0.75	12	47	20.8	9.2	2.4	8.6
U3 - cutoff = 0.85	13.5	50.5	20.7	10.2	2.3	2.8
U3 - cutoff = 0.95	16.2	53.1	19.3	8.6	1.8	1
Mean administrations						
U2 - cutoff = 0.75	6.9	12.1	5.8	2.9	0.9	-
U2 - cutoff = 0.85	7.4	12.5	5.8	3	0.8	-
U2 - cutoff = 0.95	7.7	12.8	5.8	2.8	0.8	-
U3 - cutoff = 0.75	7.2	12.6	5.4	2.6	0.7	-
U3 - cutoff = 0.85	7.7	13.2	5.4	2.6	0.7	-
U3 - cutoff = 0.95	8	13.5	5.3	2.4	0.6	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.2	4.4	1.6	0.8	0.3	-
U2 - cutoff = 0.85	1.4	4.6	1.6	0.9	0.2	-
U2 - cutoff = 0.95	0.3	3.4	1.3	0.7	0.2	-
U3 - cutoff = 0.75	1.4	4.6	1.4	0.8	0.2	-
U3 - cutoff = 0.85	1.5	4.8	1.4	0.8	0.2	-
U3 - cutoff = 0.95	0.3	3.5	1.1	0.6	0.1	-
Mean trial duration (days)						
						Days
U2 - cutoff = 0.75						1114.8
U2 - cutoff = 0.85						1151.9
U2 - cutoff = 0.95						1326.8
U3 - cutoff = 0.75						1110.9
U3 - cutoff = 0.85						1150.1
U3 - cutoff = 0.95						1341.1
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - cutoff = 0.75	15.3	8.6	3	1.4	0.3	71.4
U2 - cutoff = 0.85	28.7	12.4	3.5	1.5	0.3	53.6
U2 - cutoff = 0.95	44.7	15.3	3.4	1	0.2	35.4
U3 - cutoff = 0.75	14	9.3	3.3	1	0.1	72.3
U3 - cutoff = 0.85	28.2	14	2.4	1.3	0.2	53.9
U3 - cutoff = 0.95	45.8	15.7	1.8	0.8	0.3	35.5
Mean administrations						
U2 - cutoff = 0.75	9.4	5.4	2.2	1.1	0.3	-
U2 - cutoff = 0.85	12.6	6	2.3	1.2	0.3	-
U2 - cutoff = 0.95	15.7	6.2	2.3	1.1	0.3	-
U3 - cutoff = 0.75	9.6	5.4	2.1	1.1	0.3	-
U3 - cutoff = 0.85	12.9	5.8	2.2	1.1	0.3	-
U3 - cutoff = 0.95	16.1	5.9	2.2	1.1	0.3	-
Mean IPDE administrations						
U2 - cutoff = 0.75	2.1	2.6	0.7	0.4	0.1	-
U2 - cutoff = 0.85	3.3	2.8	0.7	0.4	0.1	-
U2 - cutoff = 0.95	0.8	2.9	0.6	0.4	0.1	-
U3 - cutoff = 0.75	2.2	2.6	0.6	0.4	0.1	-
U3 - cutoff = 0.85	3.5	2.8	0.7	0.4	0.1	-
U3 - cutoff = 0.95	0.9	2.9	0.6	0.4	0.1	-
Mean trial duration (days)						
						Days
U2 - cutoff = 0.75						683.6
U2 - cutoff = 0.85						818.9
U2 - cutoff = 0.95						1142.5
U3 - cutoff = 0.75						681.7
U3 - cutoff = 0.85						820
U3 - cutoff = 0.95						1144.4

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - cutoff = 0.75	8.9	6.1	18.1	15.9	26.4	24.6
U2 - cutoff = 0.85	10.4	11	21.2	17	27.6	12.8
U2 - cutoff = 0.95	14.9	14.9	21.8	17	26.3	5.1
U3 - cutoff = 0.75	12.2	8.6	17.5	15.1	22	24.6
U3 - cutoff = 0.85	15.4	13.2	19.8	15.1	23.3	13.2
U3 - cutoff = 0.95	20.4	18.5	17.8	15.2	22.7	5.3
Mean administrations						
U2 - cutoff = 0.75	5	4.9	5.5	5.2	5.1	-
U2 - cutoff = 0.85	6.2	5.6	6	5.4	5.2	-
U2 - cutoff = 0.95	7	6	6	5.4	5	-
U3 - cutoff = 0.75	5.2	5	5.5	5.2	4.8	-
U3 - cutoff = 0.85	6.4	5.7	5.9	5.4	4.8	-
U3 - cutoff = 0.95	7.7	5.9	5.9	5.3	4.7	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.7	3	1.8	2	1.6	-
U2 - cutoff = 0.85	1.3	3.3	1.9	2.1	1.6	-
U2 - cutoff = 0.95	0.6	3.5	1.6	1.9	1	-
U3 - cutoff = 0.75	0.8	3	1.8	2	1.5	-
U3 - cutoff = 0.85	1.4	3.3	1.9	2.1	1.5	-
U3 - cutoff = 0.95	0.6	3.5	1.6	1.8	0.9	-
Mean trial duration (days)						
U2 - cutoff = 0.75						Days
U2 - cutoff = 0.85						913.9
U2 - cutoff = 0.95						999.7
U3 - cutoff = 0.75						1154.5
U3 - cutoff = 0.85						910.2
U3 - cutoff = 0.95						997
						1167.3
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - cutoff = 0.75	2.6	28	15.6	6.1	1.8	45.9
U2 - cutoff = 0.85	4.7	36.3	15.9	6.5	1.8	34.8
U2 - cutoff = 0.95	11.2	44.6	15.4	6.7	1.4	20.7
U3 - cutoff = 0.75	3.2	30.2	14.1	5.3	0.9	46.3
U3 - cutoff = 0.85	5.2	38	15.1	5.5	1.1	35.1
U3 - cutoff = 0.95	13.1	43.8	15.7	4.6	1	21.8
Mean administrations						
U2 - cutoff = 0.75	5.9	8.5	4.6	2.2	0.6	-
U2 - cutoff = 0.85	7.5	9.7	4.8	2.2	0.7	-
U2 - cutoff = 0.95	9.6	10.7	4.7	2.2	0.6	-
U3 - cutoff = 0.75	5.9	8.9	4.4	2	0.5	-
U3 - cutoff = 0.85	7.6	10.1	4.6	2.1	0.6	-
U3 - cutoff = 0.95	9.8	11	4.5	2	0.5	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.8	3.7	1.3	0.7	0.2	-
U2 - cutoff = 0.85	1.5	4.1	1.3	0.7	0.2	-
U2 - cutoff = 0.95	0.5	4	1.1	0.7	0.2	-
U3 - cutoff = 0.75	0.8	3.8	1.2	0.6	0.2	-
U3 - cutoff = 0.85	1.5	4.3	1.2	0.7	0.2	-
U3 - cutoff = 0.95	0.6	4	1	0.6	0.1	-
Mean trial duration (days)						
U2 - cutoff = 0.75						Days
U2 - cutoff = 0.85						837.8
U2 - cutoff = 0.95						944.9
U3 - cutoff = 0.75						1183.6
U3 - cutoff = 0.85						837.2
U3 - cutoff = 0.95						944.8
						1190.2

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - cutoff = 0.75	47.2	19.1	5.1	0.7	0	27.9
U2 - cutoff = 0.85	57.3	23.4	4.4	0.6	0	14.3
U2 - cutoff = 0.95	65.1	24.1	4.7	0.5	0	5.6
U3 - cutoff = 0.75	48.6	17.7	4.9	0.5	0.1	28.2
U3 - cutoff = 0.85	59.9	21.1	4.5	0.6	0	13.9
U3 - cutoff = 0.95	69.6	20	3.8	0.7	0	5.8
Mean administrations						
U2 - cutoff = 0.75	13.7	7.3	3.3	1.4	0.2	-
U2 - cutoff = 0.85	15.4	7.8	3.4	1.4	0.2	-
U2 - cutoff = 0.95	16.5	7.8	3.3	1.4	0.2	-
U3 - cutoff = 0.75	14	7.1	3.1	1.3	0.2	-
U3 - cutoff = 0.85	15.9	7.6	3.2	1.3	0.2	-
U3 - cutoff = 0.95	17.1	7.5	3.1	1.3	0.2	-
Mean IPDE administrations						
U2 - cutoff = 0.75	3.7	3.3	1	0.5	0.1	-
U2 - cutoff = 0.85	4.4	3.5	1	0.5	0.1	-
U2 - cutoff = 0.95	1	3.4	1	0.5	0.1	-
U3 - cutoff = 0.75	3.8	3.3	0.9	0.5	0.1	-
U3 - cutoff = 0.85	4.6	3.5	0.9	0.5	0.1	-
U3 - cutoff = 0.95	1	3.3	0.9	0.4	0.1	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						950.6
U2 - cutoff = 0.85						1031.3
U2 - cutoff = 0.95						1282.1
U3 - cutoff = 0.75						945.1
U3 - cutoff = 0.85						1032.1
U3 - cutoff = 0.95						1291.3
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - cutoff = 0.75	10.9	8.8	7.9	42.2	24.3	5.9
U2 - cutoff = 0.85	10.7	10.7	8.3	43.9	24.3	2.1
U2 - cutoff = 0.95	11.4	12.6	7.8	45.4	22.5	0.3
U3 - cutoff = 0.75	13.8	11.1	7.4	40.3	21.3	6.1
U3 - cutoff = 0.85	14.5	13.6	7.1	42.4	20.4	2
U3 - cutoff = 0.95	16.6	14.9	6.6	41.2	20.3	0.3
Mean administrations						
U2 - cutoff = 0.75	4	4.7	5.2	9.4	5.5	-
U2 - cutoff = 0.85	4.4	5	5.4	9.5	5.5	-
U2 - cutoff = 0.95	4.6	5.1	5.5	9.5	5.2	-
U3 - cutoff = 0.75	4.1	4.7	5.1	9.6	5.2	-
U3 - cutoff = 0.85	4.6	5.1	5.4	9.6	5.2	-
U3 - cutoff = 0.95	4.9	5.2	5.3	9.6	4.9	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.4	3.1	1.5	3.3	1.4	-
U2 - cutoff = 0.85	0.6	3.2	1.6	3.3	1.4	-
U2 - cutoff = 0.95	0.3	3.1	1.5	2.3	1.1	-
U3 - cutoff = 0.75	0.5	3.1	1.5	3.3	1.3	-
U3 - cutoff = 0.85	0.7	3.2	1.6	3.3	1.3	-
U3 - cutoff = 0.95	0.4	3.1	1.5	2.3	1	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1049.6
U2 - cutoff = 0.85						1084
U2 - cutoff = 0.95						1190.8
U3 - cutoff = 0.75						1045.8
U3 - cutoff = 0.85						1079.6
U3 - cutoff = 0.95						1202.1

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - cutoff = 0.75	4.2	28.2	15.2	6.8	1.7	43.9
U2 - cutoff = 0.85	6.3	36.8	17.2	6.7	1.7	31.3
U2 - cutoff = 0.95	11.9	45.6	16.4	6.1	1.7	18.3
U3 - cutoff = 0.75	6.1	28.6	14.1	5.5	1.2	44.5
U3 - cutoff = 0.85	8.5	37.6	15.9	5.1	1.3	31.6
U3 - cutoff = 0.95	16.2	44.1	15.3	5.1	1.2	18.1
Mean administrations						
U2 - cutoff = 0.75	6	7.8	4.7	2.8	0.9	-
U2 - cutoff = 0.85	7.6	9.1	5	2.9	0.9	-
U2 - cutoff = 0.95	9.5	10	5	2.9	0.9	-
U3 - cutoff = 0.75	6	8.1	4.7	2.7	0.8	-
U3 - cutoff = 0.85	7.7	9.4	4.9	2.7	0.8	-
U3 - cutoff = 0.95	9.9	10.1	4.8	2.7	0.7	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.9	3.7	1.4	1	0.3	-
U2 - cutoff = 0.85	1.5	4.2	1.5	1	0.3	-
U2 - cutoff = 0.95	0.6	4.2	1.3	0.9	0.2	-
U3 - cutoff = 0.75	0.9	3.8	1.4	0.9	0.3	-
U3 - cutoff = 0.85	1.6	4.3	1.4	1	0.3	-
U3 - cutoff = 0.95	0.7	4.2	1.2	0.9	0.2	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						821.4
U2 - cutoff = 0.85						932.7
U2 - cutoff = 0.95						1158.1
U3 - cutoff = 0.75						819.7
U3 - cutoff = 0.85						930
U3 - cutoff = 0.95						1159
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - cutoff = 0.75	13.5	10.8	12.1	32.1	16	15.5
U2 - cutoff = 0.85	14.9	15.1	14.4	34.2	16	5.4
U2 - cutoff = 0.95	19.9	16.8	14.7	32.5	15.2	0.9
U3 - cutoff = 0.75	18	13.3	11.6	29.7	11.4	16
U3 - cutoff = 0.85	21.3	18.2	13.2	30.6	11.2	5.5
U3 - cutoff = 0.95	26.4	20.8	11.6	29.2	11	1
Mean administrations						
U2 - cutoff = 0.75	4.8	5.2	5.3	7.7	4.2	-
U2 - cutoff = 0.85	5.7	5.9	5.7	7.8	4.3	-
U2 - cutoff = 0.95	6.1	6.1	5.7	7.8	4.2	-
U3 - cutoff = 0.75	5	5.4	5.1	7.7	4	-
U3 - cutoff = 0.85	6	6.1	5.6	7.8	4	-
U3 - cutoff = 0.95	6.9	6.1	5.4	7.7	3.9	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.7	3.2	1.7	2.9	1.4	-
U2 - cutoff = 0.85	1.1	3.5	1.8	3	1.4	-
U2 - cutoff = 0.95	0.5	3.5	1.6	2.2	1	-
U3 - cutoff = 0.75	0.8	3.3	1.6	2.9	1.3	-
U3 - cutoff = 0.85	1.3	3.6	1.8	2.9	1.3	-
U3 - cutoff = 0.95	0.6	3.5	1.5	2.1	0.9	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						955.7
U2 - cutoff = 0.85						1028.1
U2 - cutoff = 0.95						1159.3
U3 - cutoff = 0.75						953
U3 - cutoff = 0.85						1024.9
U3 - cutoff = 0.95						1172.7

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - cutoff = 0.75	32.4	16.4	14.1	6.7	1.4	29
U2 - cutoff = 0.85	41.9	27.2	13.1	6.5	0.4	10.9
U2 - cutoff = 0.95	49.7	29.3	12.4	6.2	0.5	1.9
U3 - cutoff = 0.75	33.1	18.1	12.8	5.4	1.1	29.5
U3 - cutoff = 0.85	46.1	28.2	10	3.3	0.5	11.9
U3 - cutoff = 0.95	60.1	26.9	8.1	2.7	0.3	1.9
Mean administrations						
U2 - cutoff = 0.75	7.7	7.1	5.4	4	1.5	-
U2 - cutoff = 0.85	9.4	8	5.7	4.1	1.6	-
U2 - cutoff = 0.95	10.2	8.3	5.7	4.1	1.5	-
U3 - cutoff = 0.75	8	7.1	5.3	3.9	1.5	-
U3 - cutoff = 0.85	9.9	8	5.5	4	1.5	-
U3 - cutoff = 0.95	11.3	8	5.3	3.9	1.4	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.9	3.8	1.8	1.7	0.5	-
U2 - cutoff = 0.85	2.6	4.2	1.9	1.7	0.5	-
U2 - cutoff = 0.95	1	4.1	1.8	1.6	0.4	-
U3 - cutoff = 0.75	2	3.8	1.7	1.6	0.5	-
U3 - cutoff = 0.85	2.9	4.2	1.8	1.7	0.5	-
U3 - cutoff = 0.95	1.2	3.9	1.7	1.5	0.4	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						884.6
U2 - cutoff = 0.85						982.4
U2 - cutoff = 0.95						1149
U3 - cutoff = 0.75						883.7
U3 - cutoff = 0.85						981.6
U3 - cutoff = 0.95						1159.1
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - cutoff = 0.75	40.2	24.9	14.7	12.9	7.3	0
U2 - cutoff = 0.85	40.6	24.4	14.6	13.1	7.3	0
U2 - cutoff = 0.95	40.8	24.2	13.9	13.9	7.2	0
U3 - cutoff = 0.75	45.5	25.5	13.4	10.8	4.6	0.2
U3 - cutoff = 0.85	47.1	24.2	12.6	11	5.1	0
U3 - cutoff = 0.95	45.6	24.5	13.7	11.8	4.4	0
Mean administrations						
U2 - cutoff = 0.75	4.5	4.9	6	8.2	6.4	-
U2 - cutoff = 0.85	4.5	4.9	6	8.3	6.3	-
U2 - cutoff = 0.95	4.5	5	6.1	8.3	6.2	-
U3 - cutoff = 0.75	4.7	5.1	6.1	8.2	5.8	-
U3 - cutoff = 0.85	4.7	5.1	6.2	8.2	5.7	-
U3 - cutoff = 0.95	4.7	5.2	6.2	8.2	5.7	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.6	2.9	2.2	3.6	2.4	-
U2 - cutoff = 0.85	0.6	2.9	2.2	3.6	2.3	-
U2 - cutoff = 0.95	0.2	2.8	2	2.9	1.4	-
U3 - cutoff = 0.75	0.7	3	2.3	3.6	2.1	-
U3 - cutoff = 0.85	0.7	3	2.3	3.6	2.1	-
U3 - cutoff = 0.95	0.2	2.8	2.1	2.9	1.4	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1008.8
U2 - cutoff = 0.85						1009.6
U2 - cutoff = 0.95						1145.3
U3 - cutoff = 0.75						1008.5
U3 - cutoff = 0.85						1008.4
U3 - cutoff = 0.95						1139.8

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - cutoff = 0.75	18.7	58.5	17.2	2.4	0.1	3.1
U2 - cutoff = 0.85	18.3	61	17.4	2.8	0.1	0.4
U2 - cutoff = 0.95	20.6	60.2	16.9	2.1	0.1	0.1
U3 - cutoff = 0.75	21.6	60.1	13.5	1.9	0	2.9
U3 - cutoff = 0.85	23.2	61.2	13.2	1.9	0	0.5
U3 - cutoff = 0.95	22.5	62.7	13.2	1.5	0	0.1
Mean administrations						
U2 - cutoff = 0.75	8.8	14	5.3	1.3	0.2	-
U2 - cutoff = 0.85	9	14.1	5.4	1.3	0.2	-
U2 - cutoff = 0.95	9	14.2	5.3	1.3	0.2	-
U3 - cutoff = 0.75	9.3	14.3	4.7	1.1	0.1	-
U3 - cutoff = 0.85	9.5	14.5	4.8	1	0.1	-
U3 - cutoff = 0.95	9.6	14.7	4.6	1	0.1	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.9	4.7	1.3	0.4	0	-
U2 - cutoff = 0.85	1.9	4.7	1.4	0.4	0	-
U2 - cutoff = 0.95	0.5	3.7	1.2	0.4	0.1	-
U3 - cutoff = 0.75	2.1	4.8	1.1	0.3	0	-
U3 - cutoff = 0.85	2.1	4.8	1.2	0.3	0	-
U3 - cutoff = 0.95	0.5	3.8	1	0.3	0	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1175.4
U2 - cutoff = 0.85						1190.1
U2 - cutoff = 0.95						1330
U3 - cutoff = 0.75						1168.2
U3 - cutoff = 0.85						1183.2
U3 - cutoff = 0.95						1347.5
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - cutoff = 0.75	10.5	9.9	8.8	14.5	55.4	0.9
U2 - cutoff = 0.85	10.6	10.8	8.3	15	55.3	0
U2 - cutoff = 0.95	12.1	10.5	8.4	15.1	53.9	0
U3 - cutoff = 0.75	15.8	13.7	9.4	13	47.4	0.7
U3 - cutoff = 0.85	16.5	14.6	9.8	11.1	48	0
U3 - cutoff = 0.95	18.9	13	9.3	11.8	47	0
Mean administrations						
U2 - cutoff = 0.75	3.4	3.9	5.1	6.8	10.6	-
U2 - cutoff = 0.85	3.4	4	5.1	6.8	10.6	-
U2 - cutoff = 0.95	3.5	4	5.2	6.9	10.4	-
U3 - cutoff = 0.75	3.4	4.1	5.2	6.8	10.2	-
U3 - cutoff = 0.85	3.5	4.2	5.2	6.8	10.3	-
U3 - cutoff = 0.95	3.6	4.2	5.3	6.8	10	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.2	2.8	1.7	2.8	3	-
U2 - cutoff = 0.85	0.2	2.8	1.8	2.8	3	-
U2 - cutoff = 0.95	0.1	2.7	1.5	2.5	1.6	-
U3 - cutoff = 0.75	0.2	2.9	1.8	2.9	2.9	-
U3 - cutoff = 0.85	0.2	2.9	1.8	2.9	2.9	-
U3 - cutoff = 0.95	0.2	2.8	1.5	2.5	1.5	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1058.3
U2 - cutoff = 0.85						1067
U2 - cutoff = 0.95						1190.7
U3 - cutoff = 0.75						1053.4
U3 - cutoff = 0.85						1062.6
U3 - cutoff = 0.95						1185.9

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - cutoff = 0.75	8.5	45	18.2	2.8	0.1	25.4
U2 - cutoff = 0.85	11	53.4	19.8	3	0.1	12.7
U2 - cutoff = 0.95	15.9	59.7	18.8	2.8	0.1	2.7
U3 - cutoff = 0.75	9.6	45.7	16.2	2.1	0.1	26.3
U3 - cutoff = 0.85	13.5	55.2	16.6	2.2	0.1	12.4
U3 - cutoff = 0.95	18	60	16.6	2.4	0.1	3
Mean administrations						
U2 - cutoff = 0.75	6.1	11	6.3	2.3	0.4	-
U2 - cutoff = 0.85	7.1	12.1	6.6	2.3	0.4	-
U2 - cutoff = 0.95	8.1	12.6	6.5	2.3	0.4	-
U3 - cutoff = 0.75	6.2	11.2	6.1	2.1	0.4	-
U3 - cutoff = 0.85	7.4	12.5	6.3	2.1	0.4	-
U3 - cutoff = 0.95	8.5	12.8	6	2	0.3	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1	4.9	1.8	0.8	0.1	-
U2 - cutoff = 0.85	1.5	5.2	1.9	0.8	0.1	-
U2 - cutoff = 0.95	0.7	4.7	1.7	0.8	0.1	-
U3 - cutoff = 0.75	1	4.9	1.7	0.7	0.1	-
U3 - cutoff = 0.85	1.5	5.4	1.8	0.7	0.1	-
U3 - cutoff = 0.95	0.7	4.6	1.6	0.8	0.1	-
Mean trial duration (days)						
U2 - cutoff = 0.75						Days
U2 - cutoff = 0.85						961.2
U2 - cutoff = 0.95						1047.1
U3 - cutoff = 0.75						1202.3
U3 - cutoff = 0.85						957
U3 - cutoff = 0.95						1047.4
						1213
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - cutoff = 0.75	9.1	42.5	26.7	11.9	1.8	8
U2 - cutoff = 0.85	10.6	45.6	27.1	12.6	1.9	2.2
U2 - cutoff = 0.95	10	49.2	26.7	11.9	1.9	0.3
U3 - cutoff = 0.75	11.8	44.6	23.5	10.4	1.4	8.3
U3 - cutoff = 0.85	12.4	50.1	22.6	11.5	1.3	2.1
U3 - cutoff = 0.95	12.6	51.2	24.1	10.3	1.4	0.3
Mean administrations						
U2 - cutoff = 0.75	4.3	9.5	7.9	5.1	1.8	-
U2 - cutoff = 0.85	4.6	9.9	8.1	5.2	1.9	-
U2 - cutoff = 0.95	4.8	10.2	8.2	5	1.8	-
U3 - cutoff = 0.75	4.5	10	7.7	4.8	1.6	-
U3 - cutoff = 0.85	4.9	10.4	7.8	4.9	1.7	-
U3 - cutoff = 0.95	5.3	10.5	7.9	4.7	1.6	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.5	4.6	2.5	1.8	0.6	-
U2 - cutoff = 0.85	0.6	4.8	2.6	1.8	0.6	-
U2 - cutoff = 0.95	0.4	3.9	2.3	1.7	0.5	-
U3 - cutoff = 0.75	0.6	4.9	2.4	1.7	0.5	-
U3 - cutoff = 0.85	0.8	5	2.5	1.7	0.6	-
U3 - cutoff = 0.95	0.5	3.8	2.3	1.6	0.5	-
Mean trial duration (days)						
U2 - cutoff = 0.75						Days
U2 - cutoff = 0.85						1020.8
U2 - cutoff = 0.95						1059.4
U3 - cutoff = 0.75						1167.8
U3 - cutoff = 0.85						1016.5
U3 - cutoff = 0.95						1056.2
						1172.5

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - cutoff = 0.75	58.5	1.8	0	0	0	39.7
U2 - cutoff = 0.85	58.8	1.8	0	0	0	39.4
U2 - cutoff = 0.95	57.4	1.6	0	0	0	41
U3 - cutoff = 0.75	58.3	1.7	0	0	0	40
U3 - cutoff = 0.85	59	1.6	0	0	0	39.4
U3 - cutoff = 0.95	56.9	1.5	0	0	0	41.6
Mean administrations						
U2 - cutoff = 0.75	20.3	2.6	0.2	0	0	-
U2 - cutoff = 0.85	20.3	2.7	0.3	0	0	-
U2 - cutoff = 0.95	20.2	2.5	0.2	0	0	-
U3 - cutoff = 0.75	20.4	2.5	0.2	0	0	-
U3 - cutoff = 0.85	20.4	2.6	0.2	0	0	-
U3 - cutoff = 0.95	20.2	2.4	0.2	0	0	-
Mean IPDE administrations						
U2 - cutoff = 0.75	3.8	1	0.1	0	0	-
U2 - cutoff = 0.85	3.8	1	0.1	0	0	-
U2 - cutoff = 0.95	0.2	1.1	0.1	0	0	-
U3 - cutoff = 0.75	3.9	1	0	0	0	-
U3 - cutoff = 0.85	3.9	1	0.1	0	0	-
U3 - cutoff = 0.95	0.2	1.1	0.1	0	0	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1009.2
U2 - cutoff = 0.85						1012.1
U2 - cutoff = 0.95						1188.1
U3 - cutoff = 0.75						1006.5
U3 - cutoff = 0.85						1009.4
U3 - cutoff = 0.95						1187.8
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - cutoff = 0.75	11.7	44.8	12.2	4.3	0.1	26.9
U2 - cutoff = 0.85	14.4	56.9	13.6	4	0	11.1
U2 - cutoff = 0.95	18	64.5	12.4	3.6	0	1.5
U3 - cutoff = 0.75	12.9	45.1	11.3	3.3	0	27.4
U3 - cutoff = 0.85	16.5	57.2	11.8	3.2	0	11.3
U3 - cutoff = 0.95	21.9	62.4	10.9	3.3	0	1.5
Mean administrations						
U2 - cutoff = 0.75	6.6	11.5	5.5	2.2	0.2	-
U2 - cutoff = 0.85	7.8	12.9	5.8	2.2	0.2	-
U2 - cutoff = 0.95	8.7	13.3	5.6	2.1	0.2	-
U3 - cutoff = 0.75	6.9	11.6	5.3	2	0.2	-
U3 - cutoff = 0.85	8.2	13	5.5	2.1	0.2	-
U3 - cutoff = 0.95	9.3	13	5.4	2	0.2	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.3	5.1	1.6	0.7	0.1	-
U2 - cutoff = 0.85	1.8	5.7	1.6	0.7	0.1	-
U2 - cutoff = 0.95	0.9	4.8	1.7	0.7	0.1	-
U3 - cutoff = 0.75	1.4	5.2	1.5	0.7	0.1	-
U3 - cutoff = 0.85	2	5.7	1.6	0.7	0.1	-
U3 - cutoff = 0.95	1	4.7	1.6	0.7	0.1	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						944.3
U2 - cutoff = 0.85						1045.6
U2 - cutoff = 0.95						1199.3
U3 - cutoff = 0.75						940
U3 - cutoff = 0.85						1039.7
U3 - cutoff = 0.95						1208.2

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - cutoff = 0.75	1.9	0.3	0.1	0	0	97.7
U2 - cutoff = 0.85	8.1	2	0	0	0	89.9
U2 - cutoff = 0.95	20.5	4.8	0	0	0	74.7
U3 - cutoff = 0.75	1.9	0.4	0.2	0	0	97.5
U3 - cutoff = 0.85	8.3	1.7	0.1	0	0	89.9
U3 - cutoff = 0.95	20.8	4.4	0	0	0	74.8
Mean administrations						
U2 - cutoff = 0.75	7.7	3.9	0.7	0.1	0	-
U2 - cutoff = 0.85	11.1	4.4	0.7	0.1	0	-
U2 - cutoff = 0.95	15.8	4.8	0.7	0.1	0	-
U3 - cutoff = 0.75	7.7	3.9	0.7	0.1	0	-
U3 - cutoff = 0.85	11.1	4.4	0.7	0.1	0	-
U3 - cutoff = 0.95	15.9	4.6	0.7	0.1	0	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.4	2.1	0.2	0	0	-
U2 - cutoff = 0.85	2.9	2.3	0.2	0	0	-
U2 - cutoff = 0.95	0.7	2.8	0.2	0	0	-
U3 - cutoff = 0.75	1.4	2.1	0.2	0	0	-
U3 - cutoff = 0.85	2.8	2.3	0.2	0	0	-
U3 - cutoff = 0.95	0.7	2.6	0.2	0	0	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						466.8
U2 - cutoff = 0.85						594.5
U2 - cutoff = 0.95						971.5
U3 - cutoff = 0.75						465.5
U3 - cutoff = 0.85						591.5
U3 - cutoff = 0.95						976
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - cutoff = 0.75	2.5	2.1	3.9	53	38.3	0.2
U2 - cutoff = 0.85	2.2	2.2	3.6	54.1	37.9	0
U2 - cutoff = 0.95	2.4	1.5	3.1	57.6	35.4	0
U3 - cutoff = 0.75	3.9	3.1	5.3	56.5	31	0.2
U3 - cutoff = 0.85	4	3.1	4.8	57.1	31	0
U3 - cutoff = 0.95	3.9	3.2	4.8	59.2	28.9	0
Mean administrations						
U2 - cutoff = 0.75	3.1	3.1	3.9	12.4	7.4	-
U2 - cutoff = 0.85	3.1	3.1	3.9	12.5	7.4	-
U2 - cutoff = 0.95	3.1	3.2	3.9	12.9	6.9	-
U3 - cutoff = 0.75	3.1	3.2	4	12.9	6.7	-
U3 - cutoff = 0.85	3.1	3.2	4	13	6.7	-
U3 - cutoff = 0.95	3.1	3.2	4.1	13.3	6.3	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0	2.4	1.2	4.3	2.2	-
U2 - cutoff = 0.85	0	2.4	1.2	4.4	2.2	-
U2 - cutoff = 0.95	0	2.5	1.1	2.5	1.4	-
U3 - cutoff = 0.75	0	2.5	1.3	4.6	2	-
U3 - cutoff = 0.85	0	2.5	1.3	4.6	2	-
U3 - cutoff = 0.95	0	2.4	1.1	2.5	1.2	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1087.5
U2 - cutoff = 0.85						1090.7
U2 - cutoff = 0.95						1250.6
U3 - cutoff = 0.75						1086
U3 - cutoff = 0.85						1087.4
U3 - cutoff = 0.95						1257.4

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - cutoff = 0.75	24.5	27.2	36.5	5.6	0.1	6.1
U2 - cutoff = 0.85	25.6	29.7	37.1	6.3	0.1	1.2
U2 - cutoff = 0.95	27.9	31.6	35.1	5.3	0	0.1
U3 - cutoff = 0.75	29.3	28.4	31	4.7	0	6.6
U3 - cutoff = 0.85	31.9	29.3	31.9	5.6	0	1.3
U3 - cutoff = 0.95	34	31	30.2	4.7	0	0.1
Mean administrations						
U2 - cutoff = 0.75	8.3	8.8	8.6	2.9	0.4	-
U2 - cutoff = 0.85	8.7	9	8.8	2.9	0.4	-
U2 - cutoff = 0.95	9	9.2	8.7	2.7	0.4	-
U3 - cutoff = 0.75	8.7	9	8.5	2.6	0.3	-
U3 - cutoff = 0.85	9.4	9.1	8.5	2.5	0.3	-
U3 - cutoff = 0.95	9.6	9.3	8.3	2.5	0.3	-
Mean IPDE administrations						
U2 - cutoff = 0.75	2	3.9	2.3	0.9	0.1	-
U2 - cutoff = 0.85	2.2	4	2.4	0.9	0.1	-
U2 - cutoff = 0.95	0.7	3.7	1.9	1	0.1	-
U3 - cutoff = 0.75	2.2	4	2.2	0.8	0.1	-
U3 - cutoff = 0.85	2.5	4	2.3	0.8	0.1	-
U3 - cutoff = 0.95	0.8	3.6	1.7	0.9	0.1	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1092.8
U2 - cutoff = 0.85						1121.3
U2 - cutoff = 0.95						1248.5
U3 - cutoff = 0.75						1087.5
U3 - cutoff = 0.85						1114.8
U3 - cutoff = 0.95						1257.6
Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - cutoff = 0.75	14.9	22.1	30.7	7.4	0.1	24.8
U2 - cutoff = 0.85	18.7	29.8	32.9	7.8	0.1	10.7
U2 - cutoff = 0.95	24	32.1	32.9	7.1	0.1	3.8
U3 - cutoff = 0.75	16.6	24.1	28.3	6.2	0.1	24.7
U3 - cutoff = 0.85	21.6	28.7	31	7.3	0.1	11.3
U3 - cutoff = 0.95	29.8	31.5	29.3	5.7	0.1	3.6
Mean administrations						
U2 - cutoff = 0.75	6.7	7.9	8	3.1	0.4	-
U2 - cutoff = 0.85	8	8.6	8.3	3.2	0.4	-
U2 - cutoff = 0.95	8.9	8.9	8.3	3.1	0.4	-
U3 - cutoff = 0.75	6.9	8	7.8	3	0.4	-
U3 - cutoff = 0.85	8.3	8.7	8.2	3	0.4	-
U3 - cutoff = 0.95	9.6	8.7	8	2.9	0.4	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1.3	3.8	2.2	1.1	0.1	-
U2 - cutoff = 0.85	1.9	4.1	2.3	1.1	0.1	-
U2 - cutoff = 0.95	0.7	4	1.7	1.2	0.1	-
U3 - cutoff = 0.75	1.4	3.9	2.1	1.1	0.1	-
U3 - cutoff = 0.85	2	4.2	2.3	1	0.1	-
U3 - cutoff = 0.95	0.8	3.9	1.6	1.2	0.1	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						963.9
U2 - cutoff = 0.85						1045.7
U2 - cutoff = 0.95						1209.2
U3 - cutoff = 0.75						960.3
U3 - cutoff = 0.85						1041.9
U3 - cutoff = 0.95						1220.2

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - cutoff = 0.75	11.9	18.7	44.1	11.2	0.5	13.6
U2 - cutoff = 0.85	12.9	23.4	46.3	11.8	0.7	4.9
U2 - cutoff = 0.95	16	27.4	44.7	10.5	0.7	0.7
U3 - cutoff = 0.75	14.8	20.3	40.2	10.6	0.4	13.7
U3 - cutoff = 0.85	15.8	25.7	42.3	11	0.5	4.7
U3 - cutoff = 0.95	20.6	27.3	41.4	9.4	0.5	0.8
Mean administrations						
U2 - cutoff = 0.75	5.5	7.4	9.9	4	0.9	-
U2 - cutoff = 0.85	6.2	8	10.3	4.1	0.9	-
U2 - cutoff = 0.95	6.6	8.3	10.2	4	0.8	-
U3 - cutoff = 0.75	5.7	7.6	9.8	3.8	0.7	-
U3 - cutoff = 0.85	6.5	8.2	10.1	3.9	0.7	-
U3 - cutoff = 0.95	7.2	8.2	10	3.8	0.7	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.9	3.7	2.8	1.3	0.3	-
U2 - cutoff = 0.85	1.2	3.9	2.9	1.3	0.3	-
U2 - cutoff = 0.95	0.5	3.7	2.1	1.4	0.2	-
U3 - cutoff = 0.75	1	3.8	2.8	1.3	0.2	-
U3 - cutoff = 0.85	1.4	4	2.9	1.3	0.2	-
U3 - cutoff = 0.95	0.6	3.7	2	1.4	0.2	-
Mean trial duration (days)						
U2 - cutoff = 0.75						Days
U2 - cutoff = 0.85						1032.4
U2 - cutoff = 0.95						1087.9
U3 - cutoff = 0.75						1211
U3 - cutoff = 0.85						1029.2
U3 - cutoff = 0.95						1085.5
U3 - cutoff = 0.95						1217.8
Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - cutoff = 0.75	5.7	32.4	7.2	0.4	0	54.3
U2 - cutoff = 0.85	9	44.6	9.1	0.6	0	36.7
U2 - cutoff = 0.95	20.2	51.5	7.4	0.1	0	20.8
U3 - cutoff = 0.75	5.3	32.8	6.8	0.5	0	54.6
U3 - cutoff = 0.85	8.9	45.4	7.9	0.1	0	37.7
U3 - cutoff = 0.95	21.1	51.1	7.7	0.2	0	19.9
Mean administrations						
U2 - cutoff = 0.75	6.5	9.6	3.8	1.2	0.2	-
U2 - cutoff = 0.85	8.5	11.2	4.1	1.2	0.2	-
U2 - cutoff = 0.95	10.8	12	4	1.2	0.1	-
U3 - cutoff = 0.75	6.6	9.7	3.7	1.1	0.1	-
U3 - cutoff = 0.85	8.7	11.2	4	1.1	0.1	-
U3 - cutoff = 0.95	11.3	11.9	3.8	1.1	0.1	-
Mean IPDE administrations						
U2 - cutoff = 0.75	1	4.2	1.1	0.4	0.1	-
U2 - cutoff = 0.85	1.9	4.8	1.2	0.4	0	-
U2 - cutoff = 0.95	0.8	4.6	1.2	0.4	0.1	-
U3 - cutoff = 0.75	1.1	4.2	1.1	0.4	0	-
U3 - cutoff = 0.85	2	4.8	1.2	0.4	0	-
U3 - cutoff = 0.95	0.9	4.5	1.2	0.4	0	-
Mean trial duration (days)						
U2 - cutoff = 0.75						Days
U2 - cutoff = 0.85						794.9
U2 - cutoff = 0.95						928
U3 - cutoff = 0.75						1164.8
U3 - cutoff = 0.85						791
U3 - cutoff = 0.85						921.7
U3 - cutoff = 0.95						1171.3

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - cutoff = 0.75	61.1	29.4	4.6	0.2	0	4.7
U2 - cutoff = 0.85	63.1	30	4.4	0.2	0	2.3
U2 - cutoff = 0.95	64.9	28.4	4.2	0.2	0	2.3
U3 - cutoff = 0.75	65.3	27.1	2.8	0.2	0	4.6
U3 - cutoff = 0.85	67.2	27.3	3	0.2	0	2.3
U3 - cutoff = 0.95	67.8	27.1	2.6	0.2	0	2.4
Mean administrations						
U2 - cutoff = 0.75	17.2	8.7	2.7	0.6	0	-
U2 - cutoff = 0.85	17.3	8.9	2.7	0.6	0	-
U2 - cutoff = 0.95	17.8	8.7	2.5	0.5	0	-
U3 - cutoff = 0.75	18	8.5	2.2	0.5	0	-
U3 - cutoff = 0.85	18.2	8.6	2.2	0.5	0	-
U3 - cutoff = 0.95	18.6	8.4	2.1	0.4	0	-
Mean IPDE administrations						
U2 - cutoff = 0.75	4.2	3.1	0.7	0.2	0	-
U2 - cutoff = 0.85	4.2	3.1	0.7	0.2	0	-
U2 - cutoff = 0.95	0.8	3.1	0.9	0.2	0	-
U3 - cutoff = 0.75	4.5	3	0.6	0.1	0	-
U3 - cutoff = 0.85	4.6	3	0.6	0.1	0	-
U3 - cutoff = 0.95	0.8	2.9	0.7	0.2	0	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1163.1
U2 - cutoff = 0.85						1177.9
U2 - cutoff = 0.95						1365.8
U3 - cutoff = 0.75						1158.2
U3 - cutoff = 0.85						1172.2
U3 - cutoff = 0.95						1382.5
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - cutoff = 0.75	15.9	11.2	12.2	43.8	14.3	2.6
U2 - cutoff = 0.85	17.2	12.2	12.2	43.7	14.3	0.4
U2 - cutoff = 0.95	17.2	13.3	11.1	44.9	13.5	0
U3 - cutoff = 0.75	23.9	14.3	10.7	38.1	10.3	2.7
U3 - cutoff = 0.85	25.6	14.5	10.1	38.4	10.9	0.5
U3 - cutoff = 0.95	25.8	14.5	10.6	39.3	9.8	0
Mean administrations						
U2 - cutoff = 0.75	4.1	4.7	5.8	10.3	4.6	-
U2 - cutoff = 0.85	4.3	4.8	5.9	10.4	4.6	-
U2 - cutoff = 0.95	4.4	4.9	6	10.3	4.4	-
U3 - cutoff = 0.75	4.2	4.8	5.8	10.3	4.3	-
U3 - cutoff = 0.85	4.5	5	6	10.3	4.3	-
U3 - cutoff = 0.95	4.6	5.1	6	10.3	4	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.5	2.9	1.8	3.4	1.3	-
U2 - cutoff = 0.85	0.6	2.9	1.9	3.4	1.3	-
U2 - cutoff = 0.95	0.2	2.8	1.7	2.4	1.2	-
U3 - cutoff = 0.75	0.5	3	1.9	3.4	1.2	-
U3 - cutoff = 0.85	0.6	3	1.9	3.4	1.2	-
U3 - cutoff = 0.95	0.3	2.9	1.7	2.4	1.1	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1080.3
U2 - cutoff = 0.85						1096.3
U2 - cutoff = 0.95						1195.4
U3 - cutoff = 0.75						1075.4
U3 - cutoff = 0.85						1091.3
U3 - cutoff = 0.95						1199.6

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - cutoff = 0.75	4.8	33.8	28.2	25.6	7.5	0.1
U2 - cutoff = 0.85	5.3	34.7	27	25.9	7.1	0
U2 - cutoff = 0.95	4.7	36.9	28	24.3	6.1	0
U3 - cutoff = 0.75	7.2	40.3	27.3	20.2	4.9	0.1
U3 - cutoff = 0.85	6.8	41.2	26.4	21.2	4.4	0
U3 - cutoff = 0.95	7	41.4	26.4	20.7	4.6	0
Mean administrations						
U2 - cutoff = 0.75	3.8	10.9	7.4	5.8	2.1	-
U2 - cutoff = 0.85	3.8	11	7.4	5.9	2	-
U2 - cutoff = 0.95	3.8	11.5	7.2	5.6	1.9	-
U3 - cutoff = 0.75	3.9	11.9	7.2	5.3	1.6	-
U3 - cutoff = 0.85	3.9	11.9	7.2	5.4	1.6	-
U3 - cutoff = 0.95	3.9	12.2	7.2	5.2	1.5	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.3	4.4	2.2	1.7	0.6	-
U2 - cutoff = 0.85	0.3	4.4	2.2	1.7	0.6	-
U2 - cutoff = 0.95	0.1	2.7	1.5	1.2	0.6	-
U3 - cutoff = 0.75	0.3	4.7	2.1	1.6	0.5	-
U3 - cutoff = 0.85	0.3	4.7	2.1	1.6	0.5	-
U3 - cutoff = 0.95	0.1	2.7	1.4	1.1	0.5	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1152.6
U2 - cutoff = 0.85						1153.1
U2 - cutoff = 0.95						1327.3
U3 - cutoff = 0.75						1146.7
U3 - cutoff = 0.85						1147.9
U3 - cutoff = 0.95						1341.4
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - cutoff = 0.75	13.4	14.8	34.5	25.3	11.6	0.4
U2 - cutoff = 0.85	13.6	14.2	34.4	25.7	12	0.1
U2 - cutoff = 0.95	13.4	15	33.8	26.1	11.7	0
U3 - cutoff = 0.75	19.4	17.2	31.9	23.1	8	0.4
U3 - cutoff = 0.85	20	16.5	32.5	23.6	7.3	0.1
U3 - cutoff = 0.95	19.9	15.8	33.5	23.1	7.6	0
Mean administrations						
U2 - cutoff = 0.75	3.4	4.2	8.4	8.4	5.3	-
U2 - cutoff = 0.85	3.5	4.3	8.5	8.4	5.3	-
U2 - cutoff = 0.95	3.5	4.3	8.5	8.5	5.2	-
U3 - cutoff = 0.75	3.6	4.4	8.7	8.4	4.7	-
U3 - cutoff = 0.85	3.6	4.5	8.8	8.4	4.7	-
U3 - cutoff = 0.95	3.6	4.5	9	8.3	4.6	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.2	2.8	2.7	3.2	1.7	-
U2 - cutoff = 0.85	0.2	2.9	2.8	3.2	1.7	-
U2 - cutoff = 0.95	0.1	2.7	1.9	2.8	1.2	-
U3 - cutoff = 0.75	0.3	2.9	2.9	3.2	1.5	-
U3 - cutoff = 0.85	0.3	2.9	2.9	3.2	1.5	-
U3 - cutoff = 0.95	0.1	2.8	1.9	2.7	1	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1053.6
U2 - cutoff = 0.85						1057.6
U2 - cutoff = 0.95						1179.2
U3 - cutoff = 0.75						1050.4
U3 - cutoff = 0.85						1054.2
U3 - cutoff = 0.95						1187.9

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - cutoff = 0.75	9.2	5.8	11.5	42.6	30.4	0.5
U2 - cutoff = 0.85	7.9	5.6	12.1	43.7	30.6	0.1
U2 - cutoff = 0.95	7.9	5.4	11.7	45.2	29.8	0
U3 - cutoff = 0.75	12.7	7.5	11	42.7	25.6	0.5
U3 - cutoff = 0.85	12	7.6	11	43.6	25.7	0.1
U3 - cutoff = 0.95	12.8	7.5	11.7	44.5	23.6	0
Mean administrations						
U2 - cutoff = 0.75	3.3	3.6	5.9	10.3	6.8	-
U2 - cutoff = 0.85	3.3	3.6	5.9	10.5	6.7	-
U2 - cutoff = 0.95	3.4	3.7	6	10.4	6.5	-
U3 - cutoff = 0.75	3.3	3.7	5.9	10.5	6.3	-
U3 - cutoff = 0.85	3.4	3.8	6	10.6	6.2	-
U3 - cutoff = 0.95	3.5	3.8	6.1	10.6	6	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.1	2.6	1.8	3.7	1.8	-
U2 - cutoff = 0.85	0.1	2.7	1.8	3.7	1.8	-
U2 - cutoff = 0.95	0.1	2.6	1.5	2.8	1.2	-
U3 - cutoff = 0.75	0.1	2.7	1.8	3.7	1.7	-
U3 - cutoff = 0.85	0.2	2.7	1.8	3.8	1.6	-
U3 - cutoff = 0.95	0.1	2.7	1.5	2.7	1.1	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1085.8
U2 - cutoff = 0.85						1094.7
U2 - cutoff = 0.95						1201.8
U3 - cutoff = 0.75						1082.9
U3 - cutoff = 0.85						1090.6
U3 - cutoff = 0.95						1205.7
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - cutoff = 0.75	12.4	8.8	40.7	23.3	5.1	9.7
U2 - cutoff = 0.85	11.5	12.3	44.5	23	5.1	3.6
U2 - cutoff = 0.95	14.9	12.6	45	22.3	4.8	0.4
U3 - cutoff = 0.75	16.2	11.2	39.9	19.7	3.1	9.9
U3 - cutoff = 0.85	15.8	14.3	42.1	20.9	3.3	3.6
U3 - cutoff = 0.95	18.6	15.2	44.1	19.3	2.5	0.3
Mean administrations						
U2 - cutoff = 0.75	4.4	5.6	9.2	6.3	2.7	-
U2 - cutoff = 0.85	5	6.2	9.6	6.3	2.8	-
U2 - cutoff = 0.95	5.1	6.4	9.6	6.2	2.7	-
U3 - cutoff = 0.75	4.6	5.8	9.3	6.1	2.4	-
U3 - cutoff = 0.85	5.2	6.3	9.7	6.1	2.4	-
U3 - cutoff = 0.95	5.8	6.4	9.6	6	2.3	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.6	3.3	2.8	2.2	0.9	-
U2 - cutoff = 0.85	0.8	3.5	2.9	2.2	0.9	-
U2 - cutoff = 0.95	0.4	3.4	2.1	2	0.7	-
U3 - cutoff = 0.75	0.7	3.4	2.8	2.1	0.8	-
U3 - cutoff = 0.85	1	3.6	2.9	2.1	0.8	-
U3 - cutoff = 0.95	0.5	3.4	2	1.9	0.5	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1015.1
U2 - cutoff = 0.85						1065.3
U2 - cutoff = 0.95						1179.5
U3 - cutoff = 0.75						1009.7
U3 - cutoff = 0.85						1062.1
U3 - cutoff = 0.95						1191

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - cutoff = 0.75	9	39.5	31.8	15.7	3.8	0.2
U2 - cutoff = 0.85	9.2	40	31.6	15.2	3.9	0.1
U2 - cutoff = 0.95	9.5	41.3	31.1	14.5	3.6	0
U3 - cutoff = 0.75	12.1	46.7	27.1	10.7	3.1	0.3
U3 - cutoff = 0.85	13.6	45.6	27.4	10.6	2.7	0.1
U3 - cutoff = 0.95	13.8	46	27.3	10.3	2.6	0
Mean administrations						
U2 - cutoff = 0.75	4.3	11	8.4	4.6	1.6	-
U2 - cutoff = 0.85	4.4	11.1	8.4	4.5	1.5	-
U2 - cutoff = 0.95	4.5	11.3	8.4	4.3	1.5	-
U3 - cutoff = 0.75	4.6	12	8.1	3.9	1.3	-
U3 - cutoff = 0.85	4.6	12	8.1	4	1.3	-
U3 - cutoff = 0.95	4.5	12.3	8	3.9	1.2	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.5	4.4	2.3	1.3	0.4	-
U2 - cutoff = 0.85	0.5	4.4	2.3	1.3	0.4	-
U2 - cutoff = 0.95	0.2	3	1.9	1.2	0.3	-
U3 - cutoff = 0.75	0.6	4.8	2.2	1.2	0.4	-
U3 - cutoff = 0.85	0.6	4.8	2.2	1.2	0.3	-
U3 - cutoff = 0.95	0.2	3	1.8	1.1	0.3	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						1152.6
U2 - cutoff = 0.85						1154.1
U2 - cutoff = 0.95						1291.2
U3 - cutoff = 0.75						1145.8
U3 - cutoff = 0.85						1147.4
U3 - cutoff = 0.95						1309.2
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - cutoff = 0.75	9	30	13.8	7.6	2	37.6
U2 - cutoff = 0.85	12.2	41.6	16.1	7.9	1.7	20.5
U2 - cutoff = 0.95	20.1	49.3	15.1	7.4	1.8	6.3
U3 - cutoff = 0.75	12.3	30	12.1	6.9	1	37.7
U3 - cutoff = 0.85	16	43.1	11.6	7.6	0.8	20.9
U3 - cutoff = 0.95	24.7	49.2	11.5	6.7	1.3	6.5
Mean administrations						
U2 - cutoff = 0.75	5.9	8.3	5.2	3.5	1.2	-
U2 - cutoff = 0.85	7.5	9.8	5.6	3.6	1.2	-
U2 - cutoff = 0.95	8.8	10.5	5.5	3.5	1.1	-
U3 - cutoff = 0.75	6.1	8.5	5.1	3.3	1.1	-
U3 - cutoff = 0.85	7.8	10	5.4	3.4	1.1	-
U3 - cutoff = 0.95	9.5	10.3	5.3	3.3	1	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.9	4.2	1.6	1.3	0.4	-
U2 - cutoff = 0.85	1.6	4.8	1.7	1.3	0.4	-
U2 - cutoff = 0.95	0.7	4.7	1.6	1.1	0.3	-
U3 - cutoff = 0.75	1	4.3	1.6	1.2	0.4	-
U3 - cutoff = 0.85	1.8	4.8	1.7	1.2	0.3	-
U3 - cutoff = 0.95	0.8	4.6	1.5	1.1	0.3	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						861.8
U2 - cutoff = 0.85						976.9
U2 - cutoff = 0.95						1157.1
U3 - cutoff = 0.75						856.9
U3 - cutoff = 0.85						974.4
U3 - cutoff = 0.95						1169.4

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - cutoff = 0.75	10.2	33.7	14.3	4.1	0.5	37.2
U2 - cutoff = 0.85	12.2	48.8	16.7	4.1	0.8	17.4
U2 - cutoff = 0.95	21.1	56.2	15.2	3.4	0.5	3.6
U3 - cutoff = 0.75	11.6	34.3	11.4	3.7	0.5	38.5
U3 - cutoff = 0.85	16	50.2	13.5	2.6	0.5	17.2
U3 - cutoff = 0.95	24.4	56	12.4	3.5	0.6	3.1
Mean administrations						
U2 - cutoff = 0.75	6	9.3	5.4	3	0.9	-
U2 - cutoff = 0.85	7.4	11.2	5.8	3	0.9	-
U2 - cutoff = 0.95	8.5	11.8	5.7	2.9	0.9	-
U3 - cutoff = 0.75	6.1	9.5	5.2	2.8	0.8	-
U3 - cutoff = 0.85	7.8	11.2	5.6	2.9	0.8	-
U3 - cutoff = 0.95	9.3	11.4	5.4	2.8	0.8	-
Mean IPDE administrations						
U2 - cutoff = 0.75	0.9	4.6	1.6	1.1	0.3	-
U2 - cutoff = 0.85	1.6	5.4	1.8	1.1	0.3	-
U2 - cutoff = 0.95	0.8	5	1.6	1.1	0.3	-
U3 - cutoff = 0.75	1	4.7	1.6	1.1	0.3	-
U3 - cutoff = 0.85	1.8	5.3	1.7	1.1	0.3	-
U3 - cutoff = 0.95	1	4.9	1.6	1	0.3	-
Mean trial duration (days)						Days
U2 - cutoff = 0.75						870.6
U2 - cutoff = 0.85						993.5
U2 - cutoff = 0.95						1155.4
U3 - cutoff = 0.75						867.7
U3 - cutoff = 0.85						992.2
U3 - cutoff = 0.95						1160.3